Variational inequalities through symplectic eyes: Lagrangians, conormals, and self-duality

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Abstract

We develop a translation dictionary between variational inequalities (maximal monotone operators) and symplectic geometry, organized around two running examples: an integrable vector field, whose picture is entirely classical, and a non-integrable maximal monotone rotation, where the classical picture breaks. In the smooth finite-dimensional case "Lagrangian graph" and "monotone" are two independent axes, whose intersection is precisely the gradients of convex functions; the dual linear statement identifies Dirac structures (symmetric pairing) with skew-adjoint operators, so that both examples are Lagrangian, each for its own pairing. We record the conormal reformulation of the variational inequality as a Lagrangian intersection problem — with the unifying observation that K convex makes N_K maximal monotone, so constraint and operator are the same species on both sides of the dictionary —, Ghoussoub's self-duality through the anti-symplectic involution with its tropical Fourier reading, the identification of the proximal flow with the Hamilton–Jacobi (Hopf–Lax) flow, the Floer–Morse dictionary in the integrable case, and the volume aspect (calibrated Lagrangians, the special Lagrangian PDE, Monge–Ampère, capacities as minimax). The variational side is a mature sixty-year theory; the symplectic side of the dictionary is largely unexplored. We close with the open directions this asymmetry suggests, and a terminological guide for the three independent collisions of terminology in the area.

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1 Introduction

Alan Weinstein's creed — *everything is a Lagrangian submanifold* [43] — suggests that any reasonable mathematical object should admit a descrip-

tion in symplectic terms. The classical *variational inequality*

$$\text{find } x^* \in K \text{ such that } \langle F(x^*), y - x^* \rangle \geq 0 \quad \forall\, y \in K, \tag{1}$$

where K is convex and F a vector field (Stampacchia–Lions, 1960s), is at first sight a purely convex-analytic object. Its modern operator-theoretic home is the theory of *maximal monotone operators* (Minty, Brezis). The purpose of this note is to ask, seriously and precisely, what the creed gives back when applied here.

We carry two running examples from the start, because they separate the phenomena cleanly.

Example 1.1 (Integrable)**.** $T_{\text{int}}(x) = x$ on $\mathbb{R}$: the gradient of the convex function $\varphi(x) = x^2/2$. Smooth Lagrangian graph, monotone, the intersection of the two axes — here everything is classical.

Example 1.2 (Non-integrable)**.** $T_{\text{rot}}(x, y) = (-y, x)$ on $\mathbb{R}^2$: rotation through 90°. Maximal monotone but not a gradient, since DT is skew-symmetric; its graph is a smooth *symplectic* plane, hence not Lagrangian at any point. No potential function exists, and the classical picture breaks down exactly here.

There is a structural asymmetry worth stating upfront. The variational side is a mature sixty-year theory: existence, uniqueness, algorithms with proofs, degree theory, canonical textbooks. The symplectic side of this dictionary is, by contrast, almost entirely unexplored — Floer invariants of solution sets, deformation cohomology, capacities have no variational-side development at all. This note is an attempt to redress the balance: to import sixty years of variational knowledge into symplectic language, and to export symplectic questions back.

There is a unifying observation that organizes everything below. The two data of (1), the operator F and the constraint K, are the same species on both sides of the dictionary. Variationally: K closed convex implies its normal-cone operator N_K is maximal monotone, so both inputs are maximal monotone operators. Symplectically: $\nu_K^* = \text{gph}(N_K)$ is a (possibly singular) Lagrangian, just as $\text{gph}(F)$ is in the smooth case — and (1) is the reflected intersection of two maximal monotone graphs. The standing hypothesis "K convex" is thus not an accessory: it is exactly what puts the constraint into the same world as the operator.

The answer has several layers, and keeping them apart is the main point. In the smooth finite-dimensional setting, "Lagrangian graph" detects exactly the *gradient* case among monotone maps (Proposition 3.1). The convex constraint itself has a genuine symplectic incarnation: its

conormal bundle is Lagrangian, and (1) is equivalent to membership in it (Proposition 4.1) — hence a Lagrangian intersection problem. For general maximal monotone operators, whose graphs are *not* Lagrangian, Ghoussoub's theory of self-dual Lagrangians [24] provides what we read as a *generating function on phase space*, with self-duality expressed through an anti-symplectic involution. Through Maslov dequantization this self-duality reads as a tropical Fourier-fixed point, the tropical analogue of the Gaussian.

None of the symplectic-topological content above — Lagrangian intersection or Floer invariants of solution sets, generating-function readings of self-duality, contact-geometric friction — appears in the literature, to the best of our knowledge; see Section 14 and the literature discussion in Section 15. The two propositions below are elementary; we include proofs to fix conventions.

The translations developed in this note are summarized in Table 1; the sections that follow keep the two sides, and the limits of each translation, carefully apart.

1.1 What the program claims

This introduction is split into two parts, following the practice of Abouzaid–Blumberg [3]: the first states what the program claims and what follows from what; the second gives the proof of these applications from ingredients stated up front, with forward references to the exact sections where each ingredient is established. The reader who reads only this introduction sees the entire logical arc.

Table 1: The translation dictionary between variational inequalities and symplectic geometry.

Variational / convex-analytic side	**Symplectic side**
Solution of (1): $x^* \in K$ with $\langle F(x^*), y - x^* \rangle \geq 0$ for all $y \in K$	Conormal membership: $(x^*, -F(x^*)) \in \nu_K^*$ (Proposition 4.1)
Convex constraint K	Conormal bundle ν_K^*: a (possibly singular) Lagrangian submanifold of $T^*\mathbb{R}^n$

Variational / convex-analytic side	**Symplectic side**
K convex $\Longrightarrow$ N_K maximal monotone (Proposition 4.4): constraint and operator are the same species	$\nu_K^* = \mathrm{gph}(N_K)$: the variational inequality as the reflected intersection of two maximal monotone graphs
Gradient case $T = \partial\varphi$, φ convex	Smooth Lagrangian graph $\mathrm{gph}(T) \subset T^*\mathbb{R}^n$ (Proposition 3.1)
General maximal monotone T (graph *not* Lagrangian)	Self-dual Lagrangian L (Ghoussoub) on phase space itself, $T = \bar{\partial} L$
Skew-adjoint linear T (e.g. the rotation)	Dirac structure for the symmetric pairing (Remark 3.5)
Self-duality $L^*(p,x) = L(x,p)$	Symmetry of L under the anti-symplectic involution $\sigma(x,p) = (p,x)$
Legendre–Fenchel transform	Tropical Fourier transform (Maslov dequantization); self-duality as a tropical Fourier-fixed point
Proximal flow: Moreau envelope $e_t f$; proximal iteration $x^{k+1} = J_\lambda x^k$, the backward Euler step for $\dot{x} \in -\partial f(x)$	Hamilton–Jacobi flow $u_t + \frac{1}{2}\lvert\nabla u\rvert^2 = 0$ (Hopf–Lax) for $H(p) = \frac{1}{2}\lvert p\rvert^2$ (Theorem 7.1)
Morse homology of the constrained problem, generated by variational inequality solutions (Theorem 8.1)	Floer homology $HF_*(\mathrm{gph}(\nabla\varphi), R(\nu_K^*))$, integrable case
Volume distortion: Monge–Ampère measure $\det(D^2\varphi)\,dx$; special Lagrangian PDE $\sum_i \arctan\lambda_i = \theta$	Calibrated (volume-minimizing) Lagrangians (Harvey–Lawson, Proposition 9.1)
Minimax critical values (mountain-pass, Lusternik–Schnirelmann); the weak Minty world	Spectral invariants; Ekeland–Hofer capacities as minimax of the dual action

Variational / convex-analytic side	**Symplectic side**
Self-dual gap $\Psi_{T,K}$; its zeros are the solutions (Proposition 9.6); minimax spectrum	Gromov width of the sublevel sets $\{\Psi_{T,K} \le c\}$: it detects the free/contact transition (Proposition 9.9)
$0 \in T(\bar{x}) \iff \min_x L(x,0) = 0$	A genuine variational principle recovered from self-duality
Arnold-type bound for KKT points: $r \cdot \#\mathrm{KKT}(\varphi, K) \ge \sum_i \dim H_i(K;E)$ **?** ($r = \operatorname{rank} E$; Conjectures 6, 9)	Arnold conjecture: $(\operatorname{rank} E) \cdot \#\mathrm{Fix}(\phi) \ge \sum_i \dim H_i(M;E)$ **?** (Theorem 11.1, twisted form (9))

The program of theorems of §10 is seven statements, one line each:

(intA) Solutions of (1) are Lagrangian intersections (Theorem 10.1) [PROVED].

(intB) Euler characteristic equals degree (Theorem 10.2) [PROVED].

(intC) Floer equals Morse in the integrable case (Theorem 10.3) [CLASSICAL].

(intD2) Polyhedral Floer equals stratified Morse (Conjecture 2) [CONJECTURE] — needs G1. Nondegeneracy is meant in the explicit sense of Definition 2.2 (strict complementarity and nonsingular face Hessian at every KKT point).

(intE2) Stratified Viterbo package for conormals (Conjecture 3) [CONJECTURE] — needs G2.

(intF2) Floer spectral invariants equal finite-model spectral invariants (Conjecture 4) [CONJECTURE] — needs intD2.

(intG) A Floer-type invariant for closed maximal monotone sets (Theorem 10.19) [PROGRAM] — the maximal monotone horizon.

1.2 How each claim follows

We now derive the consequences above from the ingredients below, with a forward pointer and a status label for each. Status labels: [THEOREM] proved in this paper; [CLASSICAL] standard with citation; [COMPUTED] verified by code; [CONJECTURE] precisely stated, proof open; [OPEN] the construction itself is missing.

The dictionary (Table 1). Conormal membership (Proposition 4.1) [THEOREM, §4] and K convex $\Rightarrow N_K$ maximal monotone (Proposition 4.4) [THEOREM, §4] make (1) the reflected intersection of two maximal monotone graphs. The gradient case is detected by Lagrangian graphs (Proposition 3.1) [THEOREM, §3]. Self-duality via the anti-symplectic involution (§5) and its tropical reading (§6) are translations [THEOREM, modulo conventions].

Flows and the integrable case. The proximal/Hamilton–Jacobi identification (Theorem 7.1) [THEOREM, §7]; Floer = Morse in the integrable case (Theorem 8.1) [CLASSICAL modulo the dictionary, §8].

The program (intA)–(intG). (intA) and (intB) are proved in §10 [THEOREM]; (intC) is classical. (intD2) assembles the chain group (Proposition 10.4) [THEOREM, needs compactness hypothesis — see the erratum at Proposition 10.4] with the analytic package [CONJECTURE, G1]. (intE2) assembles per-stratum generating data (Proposition 10.8) [THEOREM, with the sign correction at Proposition 10.8] with gluing [CONJECTURE, G2]. (intF2) is conditional on (intD2) [CONJECTURE]. (intG) is a program statement [OPEN].

Capacities. The self-dual gap detects the free/contact transition (Proposition 9.9) [THEOREM, with the erratum at Proposition 9.9]; the canonical Fitzpatrick version (Proposition 9.12) [THEOREM]; the stratified exponent is conjectural [CONJECTURE, G6, with the well-posedness erratum at the Stratified exponent conjecture].

Arnold. The fixed-point form (Theorem 11.1) [CLASSICAL]; the twisted form (9) [CORRECTED FORM, see the erratum at (9)]; the variational route (§13) [PROGRAM, needs G1, G8].

Variational counterparts. The damped rotation T_{damp} (§12.1) [COMPUTED]; DCA as decategorified sheaf quantization (§12.2) [PROGRAM — one-step comparison proved; the slogan as stated needs G5; details in the companion paper [16]].

Sub-paper charter. A companion paper belongs to this cluster only if it advances a specific thread or gap of the hub: it must prove, compute, or formalize something the hub states but does not develop. Paper A [16] serves the DCA–sheaf thread (§12.2, gap G5): the dictionary, the category of DC decompositions, and the open problems the hub only names. Paper E3 [17] serves the backward-error thread: conditional backward-error certificates for DCA as a backward-exact algorithm on gauge-perturbed pairs, feeding back into the hub's DCA–sheaf dictionary. Standalone developments that do not serve a hub thread are cited

as related work, not as companions.

2 Preliminaries

The purpose of this section is to fix conventions, to define the normal-cone operator and KKT points, and to introduce the nondegeneracy hypothesis that will govern every counting statement below. As observed by Stampacchia–Lions, the variational inequality (1) is the constrained form of $0 \in F(x)$; Minty and Brezis gave it its operator-theoretic home — maximal monotonicity — which organizes everything that follows.

We work over $\mathbb{R}^n$ unless stated otherwise; the infinite-dimensional versions are standard. For $K \subset \mathbb{R}^n$ closed convex, the *normal cone* is

$$N_K(x) = \{p \in \mathbb{R}^n : \langle p, y - x\rangle \le 0 \ \forall y \in K\}, \qquad x \in K,$$

and $N_K(x) = \emptyset$ for $x \notin K$. For continuous single-valued F, (1) is equivalent to

$$0 \in F(x^*) + N_K(x^*), \tag{2}$$

see e.g. [21].

Definition 2.1 (Karush–Kuhn–Tucker)**.** For $\min_{x\in K} \varphi(x)$ with φ C^1 and $K = \{x : Ax \le b\}$ polyhedral, $x^* \in K$ is a *KKT point* if there are multipliers $\lambda \ge 0$ with

$$\nabla\varphi(x^*) + A^\top\lambda = 0, \qquad \lambda_i\,(a_i \cdot x^* - b_i) = 0 \ \ \forall i, \tag{3}$$

(stationarity, dual feasibility, complementary slackness). For general closed convex K, (3) is the single inclusion $-\nabla\varphi(x^*) \in N_K(x^*)$ — i.e. x^* solves the variational inequality (1). KKT points are thus the variational-inequality solutions in the gradient case $T = \nabla\varphi$.

Definition 2.2 (Stratified Morse index)**.** Let $K = \{x : Ax \le b\}$ be polyhedral and φ C^2. A KKT point x^* is *nondegenerate* if strict complementarity holds ($\lambda_i > 0$ for every active constraint) and $D^2\varphi(x^*)$ is nonsingular on the *critical subspace* $S(x^*) = \{d : a_i \cdot d = 0 \text{ for all active } i\}$. Its *stratified Morse index* is

$$\operatorname{ind}(x^*) = \#\{\text{negative eigenvalues of } D^2\varphi(x^*)|_{S(x^*)}\}.$$

This is the Goresky–MacPherson stratified Morse index of $\varphi|_K$ at x^* [27].

Remark 2.3 (Instantiation). In the variational inequality, the stratified Morse index counts the unstable directions that respect the active constraints. Explicitly, for $K = [0, \infty)$ and a KKT point at the boundary $x^* = 0$, the critical subspace is $S(0) = \{0\}$, so $\mathrm{ind}(0) = 0$ regardless of $\varphi''(0)$: the boundary removes the unstable direction.

A set-valued map $T : X \rightrightarrows X^*$ is *monotone* if $\langle p - q, x - y\rangle \geq 0$ for all $(x, p), (y, q)$ in its graph, and *maximal monotone* if its graph is maximal by inclusion among monotone graphs. Minty's theorem: on Hilbert space, T is maximal monotone iff $R(I + \lambda T) = X$ for some (hence all) $\lambda > 0$; the resolvent $(I + \lambda T)^{-1}$ is then single-valued and firmly nonexpansive. Standard examples: subdifferentials $\partial\varphi$ of closed convex φ (Rockafellar: these are exactly the maximal *cyclically* monotone operators), normal cones N_K, and skew-adjoint linear operators — e.g. the planar rotation $T(x, y) = (-y, x)$ is maximal monotone but not a gradient.

The *Fitzpatrick function* [22] of maximal monotone T,

$$\mathcal{F}_T(x, x^*) = \sup_{(y, y^*) \in \operatorname{gph} T} \big\{\langle x^*, y\rangle + \langle y^*, x\rangle - \langle y^*, y\rangle\big\},$$

satisfies $\mathcal{F}_T \geq \langle \cdot, \cdot \rangle$ with equality exactly on $\operatorname{gph} T$. Ghoussoub's theorem [24]: every maximal monotone T writes as $T = \bar{\partial} L$ for a convex, lower semicontinuous *self-dual Lagrangian* L on phase space $X \times X^*$, i.e. $L^*(p, x) = L(x, p)$ for the Legendre–Fenchel conjugate; then

$$0 \in T(\bar{x}) \iff \min_x L(x, 0) = 0, \tag{4}$$

a genuine minimization principle. (Here "Lagrangian" is used in the calculus-of-variations sense — a *function* — not in the symplectic sense; see Section 15.)

3 The smooth picture: Lagrangian versus monotone

The purpose of this section is to separate the two axes — Lagrangian graph and monotone operator — to prove they are independent, and to identify their intersection as the gradients of convex functions. This is Weinstein's creed [43] at the linear level: the creed predicts a symplectic description of every reasonable object, and here we find exactly where it holds and where it breaks.

Let $T^*\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$ carry $\omega((x, p), (y, q)) = \langle q, x\rangle - \langle p, y\rangle$.

Proposition 3.1. *Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be C^1. Then:*

(a) $\mathrm{gph}(T) \subset T^*\mathbb{R}^n$ *is Lagrangian iff* $DT(x)$ *is symmetric for every* x*.*

(b) T *is monotone iff* $\langle DT(x)v, v\rangle \geq 0$ *for every* x *and* v*, i.e. the symmetric part of* $DT(x)$ *is positive semidefinite.*

Consequently, on a simply connected domain,

$$\textit{Lagrangian} + \textit{monotone} \iff T = \nabla\varphi \textit{ for some convex } \varphi.$$

Proof. The tangent space to $\mathrm{gph}(T)$ at (x, Tx) is spanned by $(e_i, DT(x)e_i)$. Hence

$$\omega\big((e_i, DTe_i), (e_j, DTe_j)\big) = \langle DTe_j, e_i\rangle - \langle DTe_i, e_j\rangle,$$

which vanishes for all i, j iff $DT(x)$ is symmetric; with $\dim \mathrm{gph}(T) = n$ this is the Lagrangian condition, proving (a). For (b), monotonicity gives $\langle T(x+tv) - T(x), tv\rangle \geq 0$; dividing by t^2 and letting $t \to 0$ yields $\langle DT(x)v, v\rangle \geq 0$. Conversely, if the latter holds everywhere, then for $x \neq y$ the function $\phi(t) = \langle T(x+t(y-x)), y-x\rangle$ satisfies $\phi'(t) = \langle DT(x+t(y-x))(y-x), y-x\rangle \geq 0$, whence $\langle T(y) - T(x), y-x\rangle = \phi(1) - \phi(0) \geq 0$. If $DT(x)$ is symmetric and positive semidefinite for all x, then locally $T = \nabla\varphi$ by Poincaré's lemma and $D^2\varphi = DT \succeq 0$, i.e. φ is convex. □

Remark 3.2 (The two axes are independent). "Lagrangian" and "monotone" are independent conditions:

- Lagrangian + monotone: $T(x) = x$ (gradient of $x^2/2$);
- Lagrangian + not monotone: $T(x) = -x$ (gradient of the concave function $-x^2/2$);
- monotone + not Lagrangian: the rotation $T(x, y) = (-y, x)$;
- neither: e.g. $T(x_1, x_2) = (-x_1 + 2x_2,\ x_2)$.

In particular, "monotone $\iff$ Lagrangian graph" is false in both directions. The symplectic condition detects exactly the *gradient* case inside the monotone class.

Proposition 3.3 (Dirac duality)**.** *Let* $T : \mathbb{R}^n \to \mathbb{R}^n$ *be linear and let* $\langle (x,p), (y,q)\rangle_+ = \langle p, y\rangle + \langle q, x\rangle$ *be the symmetric pairing on* $\mathbb{R}^n \oplus \mathbb{R}^n$*. Then* $\mathrm{gph}(T)$ *is maximal isotropic for this pairing (a linear Dirac structure) if and only if* T *is skew-adjoint.*

Proof. $\langle (x, Tx), (y, Ty)\rangle_+ = \langle Tx, y\rangle + \langle Ty, x\rangle$, which vanishes for all x, y iff $\langle Tx, y\rangle = -\langle x, Ty\rangle$, i.e. $T^* = -T$. With $\dim \mathrm{gph}(T) = n$ this is maximal isotropy. □

Corollary 3.4. *A maximal monotone linear operator $T = S + A$, with $S \succeq 0$ symmetric and A skew-adjoint, is the sum of a Lagrangian-symplectic part and a Dirac part:* $\mathrm{gph}(S)$ *is Lagrangian for the symplectic pairing (Proposition 3.1) and* $\mathrm{gph}(A)$ *is Dirac for the symmetric pairing (Proposition 3.3).*

Remark 3.5 (The Dirac side of the picture)*.* Our two running examples are thus both Lagrangian, each for its own pairing — perfectly dual: Example 1.1 is Lagrangian for the symplectic pairing but not Dirac, while the rotation Example 1.2 is Dirac for the symmetric pairing but not Lagrangian-symplectic. Dirac geometry is the roof under which the two Lagrangian notions coexist: a Dirac structure is a maximal isotropic ("Lagrangian") subbundle for the symmetric pairing, involutive under the Courant bracket (automatic for constant linear Dirac structures). This is the linear shadow of a general principle: the "hard" (non-integrable, singular) cases live naturally in Dirac geometry, in the degenerate-Poisson / interconnection sense of the port-Hamiltonian school [10, 40, 41] — the framework built precisely for constrained and dissipative systems, where pure symplectic geometry breaks down.

Proposition 3.6 (Weak Minty keeps the Lagrangian)**.** *Let $T = \nabla\varphi$ be C^1 on $\mathbb{R}^n$ and satisfy the weak Minty condition $\langle F(z), z - z^* \rangle \geq \rho \|F(z)\|^2$ with $\rho < 0$ for some z^* [14] (or its pairwise form, negative comonotonicity). Then* $\mathrm{gph}(T) \subset T^*\mathbb{R}^n$ *remains Lagrangian: the relaxation degrades only the monotone axis of Proposition 3.1, not the symplectic one.*

Proof. By Proposition 3.1(a), $\mathrm{gph}(T)$ is Lagrangian iff $DT(x)$ is symmetric for every x. For $T = \nabla\varphi$, $DT = D^2\varphi$ is symmetric regardless of any Minty-type inequality; the weak Minty condition and negative comonotonicity constrain only the positive-semidefiniteness in Proposition 3.1(b). □

Remark 3.7 (Beyond monotonicity: the weak Minty world)*.* The weak Minty world keeps the Lagrangian and relaxes monotonicity (Proposition 3.6) — and it is precisely there that minimax phenomena (Section 9) can live, since mountain-pass critical points require non-convex potentials.

4 The constraint as a Lagrangian: conormal formulation

The purpose of this section is to reformulate the constraint K as the conormal Lagrangian ν_K^*, to prove the solution set is a Lagrangian intersection, and to record the unifying observation that K convex makes N_K maximal

monotone. Conormal bundles are classical; the point is that they put the constraint on the same footing as the operator, so the two data of (1) become the same species on both sides of the dictionary.

For closed convex $K \subset \mathbb{R}^n$ the *conormal bundle*

$$\nu_K^* = \{(x,p) \in T^*\mathbb{R}^n : x \in K,\ p \in N_K(x)\}$$

is a (possibly singular) Lagrangian submanifold of $T^*\mathbb{R}^n$; this is standard, see e.g. [33].

Proposition 4.1. *Let $K \subset \mathbb{R}^n$ be closed convex and $F : \mathbb{R}^n \to \mathbb{R}^n$ continuous. Then x^* solves the variational inequality* (1) *if and only if*

$$(x^*,\ -F(x^*)) \in \nu_K^*.$$

Proof. By (2), x^* solves (1) iff $-F(x^*) \in N_K(x^*)$, i.e. iff $(x^*, -F(x^*))$ lies in the conormal bundle. □

Corollary 4.2. *Let $R(x,p) = (x,-p)$ be the fiberwise reflection, an antisymplectic involution of $T^*\mathbb{R}^n$. The solution set of* (1) *is the projection to $\mathbb{R}^n$ of the intersection $\mathrm{gph}(F) \cap R(\nu_K^*)$ — a genuine Lagrangian intersection when F is (locally) a gradient, e.g. $F = \nabla\varphi$; for general F the graph need not be Lagrangian at all (see Remark 3.2).*

Proof. Proposition 4.1 says x^* solves (1) iff $(x^*, -F(x^*)) \in \nu_K^*$, i.e. iff $(x^*, F(x^*)) \in R(\nu_K^*)$. Hence the solution set is the $\mathbb{R}^n$-projection of $\mathrm{gph}(F) \cap R(\nu_K^*)$. When $F = \nabla\varphi$, both factors are Lagrangian (Proposition 3.1 for the graph; ν_K^* is always Lagrangian). □

Remark 4.3. As far as we know this observation has not appeared in print; both ingredients are standard separately.

Proposition 4.4. *Let $K \subset \mathbb{R}^n$ be nonempty closed convex. Then N_K is maximal monotone, and $\nu_K^* = \mathrm{gph}(N_K)$.*

Proof. $N_K = \partial I_K$, the subdifferential of the indicator function I_K, which is proper closed convex; subdifferentials of proper closed convex functions are maximal monotone [8]. The identity $\nu_K^* = \mathrm{gph}(N_K)$ is the definition of the conormal bundle. □

Proposition 4.5 (The conormal is always on the gradient side)**.** *Let $K \subset \mathbb{R}^n$ be nonempty closed convex. Then $N_K = \partial I_K$ is maximal* cyclically *monotone, so $\nu_K^* = \mathrm{gph}(N_K)$ always lies on the integrable (gradient) side of the 2×2 picture of Proposition 3.1, however singular it may be as a Lagrangian. By contrast, $\mathrm{gph}(F)$ for general F need not.*

Proof. $N_K = \partial I_K$ with I_K proper closed convex; subdifferentials of proper closed convex functions are maximal cyclically monotone [8]. Cyclic monotonicity implies integrability in the nonsmooth sense, placing ν_K^* on the gradient side of Proposition 3.1. □

Remark 4.6 (Constraint and operator are the same species). Proposition 4.4 is the precise form of the unifying observation in the Introduction: convexity of K is exactly what makes N_K maximal monotone, so that both data of (1) live in the same operator class, and both become (possibly singular) Lagrangians that are graphs of maximal monotone operators.

Example 4.7 (The conormal as a maximal monotone graph)**.** Let $K = [0, \infty) \subset \mathbb{R}$. Then $N_K(x) = \{0\}$ for $x > 0$, $N_K(0) = (-\infty, 0]$, and $N_K(x) = \emptyset$ for $x < 0$, so

$$\nu_K^* = \{(x, 0) : x > 0\} \cup \{(0, p) : p \leq 0\} \subset T^*\mathbb{R},$$

an L-shaped (singular) Lagrangian. Maximal monotonicity is direct: for $(x, 0)$ with $x > 0$ and $(0, p)$ with $p \leq 0$, $\langle 0 - p, x - 0\rangle = -px \geq 0$, and all other pairs give 0. With $F(x) = x$ (Example 1.1), $\mathrm{gph}(F)$ is the diagonal while $R(\nu_K^*) = \{(x, 0) : x > 0\} \cup \{(0, p) : p \geq 0\}$; they meet exactly at $(0, 0)$, projecting to the solution $x^* = 0$ of the variational inequality. For the orthant $K = \mathbb{R}^n_+$ one gets $\nu_K^* = \{(x, p) : x \geq 0,\ p \leq 0,\ \langle x, p\rangle = 0\}$ — the conormal bundle *is* the KKT complementarity set.

5 Self-duality through symplectic eyes

The purpose of this section is to translate Ghoussoub's self-duality into symplectic language, to exhibit the anti-symplectic involution behind it, and to recover the genuine variational principle (4). As observed by Ghoussoub [24], self-dual Lagrangians give variational principles for non-potential operators; we read his symmetry as geometry. The terminology "self-dual Lagrangian" for a convex *function* is due to Ghoussoub — see the terminological guide in §15.

Let X be reflexive and $X \times X^*$ carry the canonical symplectic form. The swap

$$\sigma(x, p) = (p, x)$$

satisfies $\sigma^*\omega = -\omega$: it is an anti-symplectic involution. Ghoussoub's self-duality,

$$L^*(p, x) = L(x, p), \qquad L^*(p, x) = \sup_{y,q}\{\langle p, y\rangle + \langle x, q\rangle - L(y, q)\}, \qquad (5)$$

is a symmetry of the convex function L under this involution (composed with Legendre–Fenchel duality). We read (5) as follows: where a classical Lagrangian submanifold transverse to the fibers is generated by a function S on the base via $L = \mathrm{graph}(dS)$, a general maximal monotone operator is represented by Ghoussoub's self-dual Lagrangian L on *phase space itself*, via $T = \bar{\partial} L$; self-duality is the symmetry $L^*(p, x) = L(x, p)$ of that Lagrangian function. The payoff is the genuine variational principle (4): $0 \in T(\bar{x})$ iff $\min_x L(x, 0) = 0$.

The basic example is $T = \partial\varphi$, for which $L(x, p) = \varphi(x) + \varphi^*(p)$ with φ^* the convex conjugate; then $\min_x L(x, 0) = \min_x \varphi(x) + \varphi^*(0) = 0$. Skew-adjoint operators superposed with gradients of convex functions are also covered [24].

Remark 5.1 (Terminological warning). N. Ghoussoub and L. Tzou [25] study Hamiltonian systems $-J\dot{u} = \partial H(u)$ on a symplectic space connecting "Lagrangian submanifolds" that are graphs of maximal monotone operators — but their "Lagrangian" means a zero-level set of an (anti-)self-dual Lagrangian *function* in the sense above, not a symplectic-geometric Lagrangian. The graph of a general maximal monotone operator is *not* Lagrangian for the standard symplectic form (Proposition 3.1). This collision must be flagged explicitly; see Section 15.

6 Tropical interlude: self-duality as a Fourier-fixed point

The purpose of this section is to identify the Legendre–Fenchel transform as the tropical analogue of the Fourier transform, and to read self-duality as a Fourier-fixed-point equation. This is Maslov's dequantization [32]: the dictionary gains a tropical column, and the transform $L \mapsto L^*$ becomes geometry.

The Legendre–Fenchel transform is the tropical ($\mathbb{R}_{\max}$) analogue of the Fourier transform: it is the Maslov dequantization of the Fourier–Laplace transform [32]. Writing $\mathcal{F}_{\mathrm{trop}}(L) = L^*$ for the tropical Fourier transform, self-duality (5) reads

$$\mathcal{F}_{\mathrm{trop}}(L) = L \circ \sigma^{-1},$$

i.e. L is fixed by the tropical Fourier transform up to the swap involution σ. This is the tropical analogue of the classical fact that the Gaussian $e^{-\pi x^2}$ is fixed by the (unitary) Fourier transform; indeed the "tropical Gaussian" $\varphi(x) = x^2/2$ satisfies $\varphi^* = \varphi$, and $L(x, p) = \frac{1}{2}(x^2 + p^2)$ satisfies (5). Thus self-dual Lagrangians are, in the tropical sense, the Gaussians of phase space.

7 Flows: Hamilton–Jacobi and the proximal flow

The purpose of this section is to identify the proximal flow with the Hamilton–Jacobi flow, to define the proximal iteration as the backward Euler step, and to set up the gradient-flow picture that the integrable case will then quantize. The Hopf–Lax formula is the classical bridge; we shall make use of this identification in §8.

A Hamiltonian H on $T^*\mathbb{R}^n$ generates two parallel objects: the Hamiltonian flow φ_H^t on phase space, and the Hamilton–Jacobi flow on functions,

$$u_t + H(\nabla u) = 0, \tag{6}$$

whose characteristics project the Hamiltonian trajectories. On the variational side there is a matching pair: a maximal monotone operator T generates the differential inclusion $\dot{x} \in -T(x)$ — hence the Brezis–Komura semigroup $S(t)$ — and, in the subdifferential case $T = \partial f$, the Moreau–Yosida regularization flow. The two pictures coincide.

Theorem 7.1. *Let $f : \mathbb{R}^n \to (-\infty, +\infty]$ be proper closed convex and, for $t > 0$,*

$$e_t f(x) = \inf_{y \in \mathbb{R}^n} \left\{ f(y) + \frac{|x-y|^2}{2t} \right\}$$

its Moreau envelope. Then:

(a) $u(t,x) := e_t f(x)$ is the unique viscosity solution of $u_t + \frac{1}{2}|\nabla u|^2 = 0$ on $(0,\infty) \times \mathbb{R}^n$ with $u(0,\cdot) = f$.

(b) $e_t f$ is C^1 and $\nabla e_t f = (\partial f)_t = (I - J_t)/t$, the Yosida approximation of ∂f, where $J_t = (I + t\,\partial f)^{-1}$ is the resolvent (proximal map).

(c) The proximal point iteration $x^{k+1} = J_\lambda x^k$ is the backward Euler discretization of $\dot{x} \in -\partial f(x)$.

Proof. (a) is the Hopf–Lax formula for $H(p) = \frac{1}{2}|p|^2$, whose convex conjugate is $H^*(v) = \frac{1}{2}|v|^2$: Hopf–Lax gives $u(t,x) = \min_y\{f(y) + tH^*((x-y)/t)\}$, exactly $e_t f$, the unique viscosity solution of (6); see e.g. [20] and [11]. (b) is Moreau's theorem [34]: the infimum is attained, $e_t f \in C^1$, and the gradient formula holds; see also [8]. (c): $x^{k+1} = J_\lambda x^k$ rewrites as $(x^{k+1} - x^k)/\lambda \in -\partial f(x^{k+1})$, the implicit Euler step for $\dot{x} \in -\partial f(x)$; see [8]. □

Example 7.2 (The two running examples, continued)**.** For the integrable Example 1.1, $f(x) = x^2/2$, one computes directly $e_t f(x) = x^2/(2(1+t))$, and indeed $u_t + \frac{1}{2}|\nabla u|^2 = 0$; the resolvent is $J_t x = x/(1+t)$ with $\nabla e_t f = (I - J_t)/t$. For the non-integrable Example 1.2 there is no f at all, so the Hamilton–Jacobi side of the dictionary is empty — while the Brezis–Komura semigroup of $\dot{z} \in -T_{\mathrm{rot}}(z)$ still exists (here: uniform rotation). Integrability is exactly what buys the Hamilton–Jacobi picture.

Remark 7.3. In symplectic language, the Hamiltonian $H(x,p) = \frac{1}{2}|p|^2$ generates the geodesic flow of $T^*\mathbb{R}^n$, and its Hamilton–Jacobi flow on functions *is* Moreau regularization. The dictionary entry reads: Hamiltonian flow $\longleftrightarrow$ proximal flow; time-λ map $\longleftrightarrow$ resolvent J_λ.

8 The integrable case: Floer homology is Morse homology

The purpose of this section is to prove, in the integrable case $T = \nabla\varphi$, that Floer homology is Morse homology — the one case where the dictionary is classical. Following Floer, and via PSS [36], this is the toy case against which the whole program of §10 will be measured: everything we conjecture later is the singular analogue of what is proved here.

We now prove the Floer-theoretic analogue in the integrable case, i.e. for $T = \nabla\varphi$. Put $L_0 = \mathrm{gph}(\nabla\varphi)$ and $L_1 = R(\nu_K^*)$, with $R(x,p) = (x,-p)$ as in Proposition 4.1, and assume transverse intersection.

Theorem 8.1. *Let $\varphi : \mathbb{R}^n \to \mathbb{R}$ be smooth and Morse, $K \subset \mathbb{R}^n$ closed convex, and L_0, L_1 as above.*

(a) The Floer chain group $CF_*(L_0, L_1)$ is freely generated by the solutions of the variational inequality for $(\nabla\varphi, K)$.

(b) For $K = \mathbb{R}^n$ (so L_1 is the zero section), there is a canonical grading-preserving isomorphism

$$HF_*(L_0, L_1) \;\cong\; HM_*(\varphi),$$

the Morse homology of φ: the Maslov index of an intersection point equals the Morse index of the corresponding critical point.

Proof. (a) is Proposition 4.1: intersection points $(x, \nabla\varphi(x)) \in L_0 \cap L_1$ are exactly the points with $-\nabla\varphi(x) \in N_K(x)$, i.e. the solutions of the variational inequality. (b) is the PSS isomorphism [36] (see also [2] for the cotangent-bundle Lagrangian version): for the pair $(\mathrm{gph}(\nabla\varphi), \text{zero section})$ the Lagrangian Floer complex is the Morse complex of φ. □

9 Volume: calibrated Lagrangians and Monge–Ampère

The purpose of this section is to develop the volume aspect of the dictionary: calibrated Lagrangians, the special Lagrangian PDE, and capacities as minimax quantities. As Harvey–Lawson taught, calibrations detect volume-minimizers; here the capacity profile will detect the free/contact transition. We begin by elaborating on the minimax picture sketched in the introduction.

The symplectic volume form $\omega^n/n!$ raises the question of "size" in our dictionary. On the symplectic side this is calibrated geometry: a Lagrangian $L \subset \mathbb{C}^n$ is *special* of phase θ if $\mathrm{Im}(e^{-i\theta}\Omega)|_L = 0$, where $\Omega = dz_1 \wedge \cdots \wedge dz_n$; Harvey–Lawson [29] prove that special Lagrangians are calibrated by $\mathrm{Re}(e^{-i\theta}\Omega)$, hence volume-minimizing under compactly supported variations. For gradient graphs this becomes a PDE:

Proposition 9.1. *Let $\varphi : \mathbb{R}^n \to \mathbb{R}$ be smooth and $L = \mathrm{gph}(\nabla\varphi) \subset T^*\mathbb{R}^n \cong \mathbb{C}^n$. Then L is special Lagrangian of phase θ if and only if*

$$\sum_{i=1}^{n} \arctan(\lambda_i(x)) = \theta \quad \forall x,$$

where $\lambda_i(x)$ are the eigenvalues of $D^2\varphi(x)$.

Proof. Write $z_j = x_j + iy_j$ with $y = \nabla\varphi(x)$ on L. Then $dz_j|_L = dx_j + i\sum_k \varphi_{jk}\, dx_k$, whence $\Omega|_L = \det(I + iD^2\varphi)\, dx_1 \wedge \cdots \wedge dx_n$. Since $D^2\varphi$ is symmetric, $\det(I+iD^2\varphi) = \prod_{i=1}^n (1+i\lambda_i)$, whose argument is $\sum_i \arctan\lambda_i$. The special Lagrangian condition $\mathrm{Im}(e^{-i\theta}\Omega|_L) = 0$ is exactly that this argument equals θ. □

Example 9.2 (The integrable example is volume-minimizing)**.** For Example 1.1, $\varphi(x) = |x|^2/2$, $D^2\varphi = I$ and $\sum_i \arctan(1) = n\pi/4$: the graph of T_{int} is special Lagrangian of phase $n\pi/4$, hence calibrated. The rotation Example 1.2 is not Lagrangian at all, so the calibrated story does not apply — another facet of the hard case.

Remark 9.3 (Monge–Ampère and optimal transport). The "volume distortion" of the gradient map is the Monge–Ampère measure $\det(D^2\varphi)\, dx$ (Aleksandrov sense for nonsmooth φ); Brenier's theorem [9] identifies $\nabla\varphi$, φ convex, as the optimal transport map for quadratic cost. The volume entry of the dictionary thus reads: calibrated (volume-minimizing) Lagrangians $\longleftrightarrow$ the special Lagrangian PDE $\sum_i \arctan\lambda_i = \theta$, a fully nonlinear PDE of Monge–Ampère type, with the Monge–Ampère measure as its volume-distortion side.

Remark 9.4 (Capacities are minimax). The Ekeland–Hofer capacities are minimax values of Clarke's dual action functional [19] — a convex-analytic object: "symplectic size" is already, classically, a minimax quantity. Its finite-dimensional ancestor is Lusternik–Schnirelmann / mountain-pass theory [1]: a sequence of minimax critical values $c_1 \leq c_2 \leq \cdots$, mirroring the Ekeland–Hofer sequence. Hence the dictionary entry, in the integrable case: spectral invariants / capacities $\longleftrightarrow$ minimax critical values. Note the mountain pass needs non-convex φ — i.e. non-monotone F — so this entry naturally lives in the weak Minty world (Remark 3.7), beyond classical monotone theory.

9.1 A worked example: capacity detects the free/contact transition

We work out the capacities direction completely in a one-parameter example, via the self-dual gap functional. Let T be maximal monotone with self-dual Lagrangian L ($T = \bar{\partial} L$, [24]) and K closed convex, with support function σ_K and indicator I_K.

Definition 9.5 (Self-dual gap functional)**.**

$$\Psi_{T,K}(x,p) = \big[L(x,p) - \langle x,p\rangle\big] + I_K(x) + \big[\sigma_K(-p) + \langle p,x\rangle\big], \qquad (x,p) \in X \times X^*.$$

Each bracket is nonnegative: $L \geq \langle \cdot,\cdot\rangle$ by Fenchel–Young and self-duality $L^*(p,x) = L(x,p)$; $I_K \geq 0$; $\sigma_K(q) \geq \langle q,x\rangle$ for $x \in K$.

Proposition 9.6. *$\Psi_{T,K}(x,p) = 0$ if and only if $x \in K$, $p \in T(x)$ and $-p \in N_K(x)$; i.e. the zeros of $\Psi_{T,K}$ project exactly onto the solution set of $0 \in T(x) + N_K(x)$.*

Proof. $L(x,p) = \langle x,p\rangle$ iff $p \in \bar{\partial} L(x) = T(x)$, by definition of the self-dual vector field; $I_K(x) = 0$ iff $x \in K$; and for $x \in K$, $\sigma_K(-p) + \langle p,x\rangle = 0$ iff $\langle -p,x\rangle = \max_{y\in K}\langle -p,y\rangle$ iff $-p \in N_K(x)$. □

Remark 9.7. For $K = X$ (unconstrained), $\sigma_X(q) = 0$ if $q = 0$ and $+\infty$ otherwise, so $\Psi_{T,X}(x,0) = L(x,0)$: this recovers Ghoussoub's principle $\min_x L(x,0) = 0$ (4).

Example 9.8 (The free/contact transition)**.** Let $T_a(x) = x - a$ on $\mathbb{R}$ ($a \in \mathbb{R}$ a parameter) and $K = [0,\infty)$. The variational inequality $x^* \geq 0$, $(x^* - a)(y - x^*) \geq 0$ for all $y \geq 0$ has the unique solution $x^*(a) = \max(a,0)$: stuck at 0 (contact) for $a < 0$, sliding with a (free) for $a > 0$; the solution map is nonsmooth at $a = 0$.

Here $\varphi_a(x) = x^2/2 - ax$, $L_a(x,p) = x^2/2 - ax + (p+a)^2/2$, and a direct computation gives, for $x \geq 0$, $p \geq 0$,

$$\Psi_a(x,p) = \tfrac{1}{2}\big[(x-p) - a\big]^2 + px,$$

and $+\infty$ otherwise. The unique zero is $(\max(a,0), \max(-a,0))$: it tracks the solution, while $p^*(a) = \max(-a,0)$ is the contact force, with the complementarity $x^*(a)\,p^*(a) = 0$.

Proposition 9.9. *Let $w_a(c) = c_G(\{\Psi_a \leq c\})$ be the Gromov width of the sublevel set in Example 9.8. For $c > 0$ small,*

$$w_a(c) = \frac{\pi}{4}\big(\sqrt{a^2+2c} - |a|\big)^2 \quad (a \neq 0), \qquad w_0(c) = \frac{2\pi c}{3+2\sqrt{2}} \quad (a = 0).$$

In particular $w_a(c)/c \to 0$ for $a \neq 0$ while $w_0(c)/c \to 2\pi/(3+2\sqrt{2}) > 0$: the asymptotic order of the capacity jumps exactly at the free/contact transition $a = 0$.

Sketch. For $a > 0$, $\{\Psi_a \leq c\} = \{(x,p) : x \geq 0,\ p \geq 0,\ (x-a)^2 + (p+a)^2 \leq a^2 + 2c\}$; it lies in the strip $0 \leq p \leq \sqrt{a^2+2c} - a$, so any embedded ball has radius at most $(\sqrt{a^2+2c} - a)/2$, and the ball centered at (a, r_0) with $r_0 = (\sqrt{a^2+2c} - a)/2$ attains it. The case $a < 0$ is analogous. For $a = 0$ the sublevel is the quarter-disk of radius $\sqrt{2c}$, whose largest inscribed ball has radius $\sqrt{2c}/(1+\sqrt{2})$. □

The self-dual gap above needs a self-dual representer L, which does not exist canonically in general. The Fitzpatrick function is the canonical replacement: for maximal monotone T,

$$F_T(x,p) = \sup_{(y,q) \in \operatorname{gph} T} \{\langle x, q\rangle + \langle y, p\rangle - \langle y, q\rangle\}$$

is the pointwise minimum of the Fitzpatrick family $\{h \text{ closed convex} : h \geq \langle\cdot,\cdot\rangle,\ h = \langle\cdot,\cdot\rangle \text{ on } \operatorname{gph} T\}$ (Burachik–Svaiter) — canonical, no choices.

Definition 9.10 (Fitzpatrick gap functional)**.**

$$\Psi^F_{T,K}(x,p) = \big[F_T(x,p) - \langle x,p\rangle\big] + I_K(x) + \big[\sigma_K(-p) + \langle p,x\rangle\big], \qquad (x,p) \in X \times X^*.$$

Each bracket is nonnegative. Put $w^F_{T,K}(c) = c_G(\{\Psi^F_{T,K} \leq c\})$ for the capacity profile; for any other representer h in the Fitzpatrick family, $F_T \leq h$ gives $w^F(c) \geq w^h(c)$ for all c, so the Fitzpatrick profile is the universal upper bound over representers.

Remark 9.11 (Instantiation). Explicitly, $w^F_{T,K}(c)$ is the Gromov width — the supremum of πr^2 over symplectically embedded balls — of the sublevel set $\{\Psi^F_{T,K} \le c\}$. As $c \to 0^+$ it probes the infinitesimal shape of the solution set: the exponent of $w^F(c)$ detects how the zero set branches at the free/contact transition.

Proposition 9.12 (Fitzpatrick gap zeros; proved)**.** *$\Psi^F_{T,K}(x,p) = 0$ if and only if $x \in K$, $p \in T(x)$ and $-p \in N_K(x)$; i.e. the zeros of $\Psi^F_{T,K}$ project exactly onto the solution set of $0 \in T(x) + N_K(x)$. No self-duality is used.*

Proof sketch. $F_T(x,p) = \langle x, p\rangle$ iff $(x,p) \in \operatorname{gph} T$, by the Fitzpatrick representer property (maximality of T); the K-bracket vanishes iff $-p \in N_K(x)$ as in Proposition 9.6. Self-duality belongs to Ghoussoub's minimizer machinery, not to the zero characterization. □

Example 9.13 (Fitzpatrick gap for the running example)**.** For $T_a(x) = x - a$, $K = [0,\infty)$: $F_{T_a}(x,p) = (x+p+a)^2/4 - ax$ and, on the first quadrant, $\Psi^F_a(x,p) = ((x-p)-a)^2/4 + xp$, with zeros exactly the variational-inequality solutions. A direct computation gives

$$w^F_0(c) = 2c \quad \text{exactly (the triangle } \{x+p \le 2\sqrt{c}\}), \qquad w^F_a(c) = \frac{8c^{3/2}}{3|a|} + O(c^{5/2}) \quad (a \neq$$

so the small-c exponent jumps $1 \to 3/2$ at the free/contact transition $a = 0$, canonically — independent of any choice of representer.

Conjecture 1 (Stratified exponent)**.** *Under a tame normal form for the zero set of (T,K) — weaker than full o-minimal definability —* and *assuming the sublevel sets $\{\Psi^F_{T,K} \le c\}$ have nonempty interior with $0 < w^F_{T,K}(c) < \infty$ for small $c > 0$ (with suitable localization at the zero set), the limit $\alpha = \lim_{c\to 0^+} \log w^F_{T,K}(c)/\log c$ exists and is computed from stratified normal forms of the zero set: each stratum contributes a volume exponent e_j and a width exponent a_j, with $\alpha = \min_j a_j$. In dimension one the normal forms are verified: Morse point $e = 1$, fold $e = 3/2$, transverse segment $e = 1$; the smooth case recovers the complementarity-pattern count. Nonsmoothness does not remove the exponent — it stratifies the zero set (e.g. $T = \partial|\cdot|$ gives $\alpha = 1$ where the naive count predicts $3/2$).*

Remark 9.14. Definability is *sufficient* for each fixed (T,K), not necessary: two non-definable examples (a log-periodic T; a φ with kinks accumulating at 2^{-n}) still give a stable limit $\alpha = 1$. Definability remains needed for uniform control over families. The oscillating-power obstruction is open: no fixed maximal monotone T is known for which the limit fails to exist.

10 A program of theorems: Lagrangian intersection invariants of solution sets

The purpose of this section is to state, as seven theorems, what a full Lagrangian-intersection theory of solution sets would give. We begin by elaborating on the program sketched in the introduction: each statement is given with its proof status, and the remarks say exactly what remains. This is Floer's program transplanted — he built homology from gradient-flow trajectories; we ask for homology from variational-inequality solution sets.

Proposition 4.1 identifies the solution set of (1) with a Lagrangian intersection. We formulate what a full intersection theory would give as a series of theorems, marking the status of each.

Theorem 10.1 (Solutions as intersections; proved)**.** *Let $K \subset \mathbb{R}^n$ be closed convex and F continuous maximal monotone. With $L_0 = \mathrm{gph}(F)$ and $L_1 = R(\nu_K^*)$, x^* solves the variational inequality for (F, K) if and only if $(x^*, F(x^*)) \in L_0 \cap L_1$; the solution set is the base projection of the reflected intersection of two maximal monotone graphs (Propositions 4.1 and 4.4).*

Proof. The point $(x^*, F(x^*))$ always lies in $L_0 = \mathrm{gph}(F)$. By Proposition 4.1, $\nu_K^* = \mathrm{gph}(N_K)$, so $L_1 = R(\nu_K^*) = \{(x, -p) : p \in N_K(x)\}$. Hence $(x^*, F(x^*)) \in L_1$ if and only if $-F(x^*) \in N_K(x^*)$, i.e. $0 \in F(x^*) + N_K(x^*)$, which is the variational inequality for (F, K) by (2). The solution set is thus the base projection $\pi_x(L_0 \cap L_1)$: the reflected intersection of the two maximal monotone graphs (maximality from Proposition 4.4). □

Theorem 10.2 (Euler characteristic equals degree; proved)**.** *Let φ be Morse with $\nabla\varphi$ proper, $K = \mathbb{R}^n$, $L_0 = \mathrm{gph}(\nabla\varphi)$ and $L_1 = R(\nu_{\mathbb{R}^n}^*)$ the zero section. Then*

$$\chi(CF_*(L_0, L_1)) = \sum_{x^* \text{ critical}} (-1)^{\mathrm{ind}(x^*)} = \chi(HM_*(\varphi)) = \deg(\nabla\varphi),$$

the Brouwer degree. If K is a convex polyhedron and F is transverse to the strata of K, the signed count $\#(L_0 \cap R(\nu_K^*))$ equals Robinson's normal-map degree [37].

Proof. For $K = \mathbb{R}^n$, $R(\nu_{\mathbb{R}^n}^*)$ is the zero section and $CF_*(L_0, L_1)$ is the classical exact Floer complex; Theorem 10.3 gives $\chi(CF_*) = \chi(HM_*(\varphi))$, and Poincaré–Hopf identifies the latter with $\deg(\nabla\varphi)$. For polyhedral K under transversality, $L_0 \pitchfork R(\nu_K^*)$ is a finite signed set, and the signed intersection number is the mapping degree of the normal map by classical intersection theory. □

Theorem 10.3 (Floer equals Morse; classical)**.** *For $K = \mathbb{R}^n$, $HF_*(L_0, L_1) \cong HM_*(\varphi)$ canonically (Theorem 8.1; PSS [36]).*

Proof. This is Theorem 8.1(b): for $K = \mathbb{R}^n$ the conormal ν_K^* is the zero section, so $L_1 = R(\nu_K^*)$ is the zero section, and the PSS isomorphism [36] identifies $HF_*(L_0, L_1)$ with $HM_*(\varphi)$ canonically, matching Maslov and Morse indices. □

Proposition 10.4 (The candidate Floer chain group; proved)**.** *Let $K \subset \mathbb{R}^n$ be a compact convex polyhedron and φ with every KKT point nondegenerate in the explicit sense of Definition 2.2 (strict complementarity and nonsingular face Hessian) and $\nabla\varphi$ transverse to the strata of K. Then $\mathrm{gph}(\nabla\varphi) \pitchfork R(\nu_K^*)$ in finitely many points, canonically in bijection with the KKT points — equivalently the variational inequality solutions — and*

$$CF_* = \bigoplus_{x^* \ KKT} \mathbb{Z}_2 \cdot x^*$$

is a well-defined graded vector space for the stratified Morse index (Definition 2.2).

Proof. The bijection is Propositions 4.1 and 4.4; transversality makes the intersection finite and transverse. The stratified Morse index is Definition 2.2. □

Remark 10.5 (Forward pointer). This compactness hypothesis will be essential everywhere in §10 where finiteness is drawn from transversality; we shall invoke it without further comment.

Conjecture 2. *In the setting of Proposition 10.4, for generic compatible J the moduli spaces of J-holomorphic strips with boundary on $(\mathrm{gph}(\nabla\varphi),\ R(\nu_K^*))$ are smooth compact manifolds with corners, $\partial^2 = 0$, the resulting Floer homology $HF_*(\mathrm{gph}(\nabla\varphi), R(\nu_K^*))$ is independent of choices, and it is canonically isomorphic to the stratified Morse homology of $\varphi|_K$ [27], with the Floer grading matching the stratified Morse index.*

Theorem 10.6 (Sheaf-theoretic Floer homology)**.** *Let $K \subset \mathbb{R}^n$ be a convex polyhedron and $\varphi : \mathbb{R}^n \to \mathbb{R}$ real-analytic, satisfying stratified transversality: at every KKT point, $-\nabla\varphi$ lies in the relative interior of N_K and the restricted Hessian is nondegenerate on the critical subspace. Put $R(x, p) = (x, -p)$. Then:*

(a) The sheaf-theoretic Floer algebra $HF^*_{\mathrm{sheaf}}(\varphi, K) := \mathrm{Ext}^*(F_\varphi, \mathbb{D}\mathbb{C}_K)$ is well-defined and finite-dimensional over $\mathbf{F}_2$, where $\mathbb{D}\mathbb{C}_K$ is the Verdier dual of the constant sheaf on K and F_φ is the GKS sheaf of the fiber translation $(x, \xi) \mapsto (x, \xi + d\varphi_x)$.

(b) *With a subanalytic stop Λ at infinity, $HW^*(L_\varphi, L_K) \cong \mathrm{Ext}^*(F_\varphi, \mathbb{D}\mathbb{C}_K)$ in the stopped wrapped Fukaya category.*

(c) *There is a canonical isomorphism $HF^*_{\mathrm{sheaf}}(\varphi, K) \cong HM^*_{\mathrm{strat}}(\varphi|_K; \mathbf{F}_2)$ (stratified Morse homology), with Floer degree equal to the stratified Morse index.*

The real-analytic hypothesis is strictly stronger than the smooth hypothesis of Conjecture 2; it is needed for GKS quantization and the subanalytic stop at infinity. Two items remain: (O1) extracting the finite version from the wrapped level — partially resolved: the differential is definable (the fiber product through the kink is fine away from the touch value, and avoiding it is generic), leaving continuation and compactness at the touch point [Open]; (O2) canonicity under the choices of stop and smoothing [Open].

Theorem 10.7 (Branch selection at junctions)**.** *Let x_- be a junction point of $R(\nu_K^*)$ where exactly two smooth branches B_1, B_2 meet cleanly (in the symplectic normal slice, two Lagrangian lines at angle $\pi/2$). Let u be a J-holomorphic strip with $u(s,1) \in B_1 \cup B_2$ for all large s. Then the junction foot $u(s,1)$ never changes branches: it lies entirely in B_1 or entirely in B_2. The selector is the sign of the detector $\varphi(s) = u(s,1)^2$, a real-analytic function whose sign is constant; the sign determines the branch. The same holds coordinate-wise for two transverse Lagrangian n-planes in $\mathbb{C}^n$, and for the $2D$ orthant corner via a product trick with complementary linear forms. In particular the junction hypothesis (J1) for the linearized b-calculus at strip ends is now satisfied.*

Proof sketch. Put $v = u^2$, holomorphic. Since $w \in \mathbb{R} \cup i\mathbb{R}$ implies $w^2 \in \mathbb{R}$, the detector $\varphi(s) = v(s,1) = u(s,1)^2$ is real-analytic and real-valued. If φ changed sign, let τ be the first sign-change zero; τ is finite, constructed from a sign-changing pair, so no compactification of the strip end is needed. Then $\mathrm{ord}_\tau(\varphi) = 2m'$ is even (squaring doubles the vanishing order of the holomorphic u), but a sign-changing zero of a real-analytic function has odd order — contradiction. Hence φ keeps its sign; $\varphi \geq 0$ forces $u(s,1) \in \mathbb{R}$ and $\varphi \leq 0$ forces $u(s,1) \in i\mathbb{R}$. □

Proposition 10.8 (Stratified generating data for conormals; proved)**.** *Let $K \subset \mathbb{R}^n$ be closed convex. Then the microsupport $SS(\mathbb{C}_K)$ of the constant sheaf is the* reflected *normal graph $R(\mathrm{gph}\, N_K)$, where $R(x,p) = (x,-p)$ is the fiberwise reflection [30]; with the outward-normal convention in use here, $SS(\mathbb{C}_K)$ carries positive covectors at the boundary (e.g. for $K = [0,\infty)$, $N_K(0) = (-\infty, 0]$ but $SS(\mathbb{C}_K)$ has $p \geq 0$ at 0). For K polyhedral/subanalytic the reflected graph is the correct identification; convexity*

alone does not imply constructibility or a sheaf-quantization/Fukaya correspondence. Via sheaf quantization [26, 35], $R(\nu_K^*)$ hence carries canonical generating data. For K polyhedral the local models are canonical stratified generating data: the piecewise-quadratic $C^{1,1}$ function S_K on the projection-cell decomposition (class $\mathcal{C}_{\mathrm{strat}}$).

Proof sketch. $SS(\mathbb{C}_K) = R(\mathrm{gph}\, N_K)$ is Kashiwara–Schapira [30] (the reflection corrects the sign in v0.31, which wrote $SS(\mathbb{C}_K) = \nu_K^*$); the quantization turning sheaves into Fukaya data is [26, 35]. Near each face F of a polyhedron, $C(R(\nu_K^*))$ restricts to the graph of dS_F with S_F quadratic on the projection cell Q_F; no fiber variables are needed. □

Conjecture 3. *The conormal ν_K^* of a polyhedral K is the disjoint union $\bigsqcup_F L_F$ over the face lattice, with $L_F = \mathrm{relint}(F) \times N_K(F)$ smooth Lagrangians; all singularity lives in the gluing at face junctions. It carries the canonical stratified generating datum $\mathcal{D}_K$ of Proposition 10.8 in the class $\mathcal{C}_{\mathrm{strat}}$ (piecewise-quadratic $C^{1,1}$, conical at infinity, 2-homogeneous principal parts), and:*

(E2-i) Existence and rigidity. *$\mathcal{D}_K$ exists; every admissible stratified generating datum is equivalent to $\mathcal{D}_K$ up to the moves (M1)–(M4): per-stratum stabilization, per-stratum homogeneous fiber diffeomorphisms, cell-subdivision refinement, and change of Euclidean metric. Here "admissible" includes the fiber-homogeneity axiom (T2.6).*

(E2-ii) Moves and GF-homology invariance. *There is a well-defined stratified GF homology $HM_*^{\mathrm{strat}}(\mathcal{D})$, invariant under (M1)–(M4), with $HM_*^{\mathrm{strat}}(\mathcal{D}_K) \cong H_*(K; \mathbf{F}_2)$.*

(E2-iii) Comparison with Floer. *$HM_*^{\mathrm{strat}}(\mathcal{D}_K)$ coincides with the Floer homology of Conjecture 2, via desingularized PSS along the branched covers.*

We write "Viterbo-type" rather than Viterbo's literal quadratic-at-infinity class: the conormal has infinite volume and is conical at infinity, so the quadratic class is the wrong regularity (the conormal is not compactly supported Hamiltonian-isotopic to the zero section). The stratified conical class $\mathcal{C}_{\mathrm{strat}}$ is the correct one; the subtlety is exactly the infinity data.

Proposition 10.9 (No single classical generating function)**.** *No C^1 function $F : \mathbb{R} \times \mathbb{R}^N \to \mathbb{R}$ with 0 a regular value of $\partial_\xi F$ generates the conormal $\nu_{[0,\infty)}^*$ exactly.*

Proof. If 0 is a regular value of $\partial_\xi F$, the generated set $\Gamma_F = \{(x, \partial_x F(x,\xi)) : \partial_\xi F(x,\xi) = 0\}$ is a smooth Lagrangian immersion, by the standard generating-function construction. But $\nu^*_{[0,\infty)} = \{(x,0) : x > 0\} \cup \{(0,p) : p \le 0\}$ is singular at the origin (an L-shaped corner), hence cannot coincide with any such Γ_F. □

10.1 Example: stratified generating data for the orthant

We work out the stratified generating datum $\mathcal{D}_K$ of Conjecture 3 completely for the running 2-dimensional example $K = [0,\infty)^2$. This is the first case where the singular conormal has a genuine corner stratum, and every formula below is exact.

The face lattice has four faces. Write points of $T^*\mathbb{R}^2$ as (x_1, x_2, p_1, p_2).

Proposition 10.10 (Strata of ν^*_K)**.** *With $L_F = \mathrm{relint}(F) \times \mathrm{ri}(N_K(F))$,*

$$\nu^*_K = L_{\mathrm{int}} \sqcup L_1 \sqcup L_2 \sqcup L_0 \sqcup R_1^+ \sqcup R_1^- \sqcup R_2^+ \sqcup R_2^- \sqcup L_{00},$$

a disjoint union of nine smooth strata: four 2-dimensional Lagrangians, four 1-dimensional isotropic rays (the boundary rays of the two facets, where one covector coordinate vanishes), and the 0-dimensional origin stratum:

$$\begin{aligned}
L_{\mathrm{int}} &= \{(x_1,x_2,0,0) : x_1 > 0,\ x_2 > 0\},\\
L_1 &= \{(0,x_2,-a,0) : x_2 > 0,\ a > 0\},\\
L_2 &= \{(x_1,0,0,-b) : x_1 > 0,\ b > 0\},\\
L_0 &= \{(0,0,-a,-b) : a > 0,\ b > 0\},\\
R_1^+ &= \{(0,x_2,0,0) : x_2 > 0\}, \quad R_1^- = \{(0,0,-a,0) : a > 0\},\\
R_2^+ &= \{(x_1,0,0,0) : x_1 > 0\}, \quad R_2^- = \{(0,0,0,-b) : b > 0\},\\
L_{00} &= \{(0,0,0,0)\},
\end{aligned}$$

for the faces $F_{\mathrm{int}} = \{x_1 > 0, x_2 > 0\}$, $F_1 = \{x_1 = 0, x_2 > 0\}$, $F_2 = \{x_2 = 0, x_1 > 0\}$, $F_0 = \{(0,0)\}$: $4 + 4 + 1 = 9$.

Proof. For $K = [0,\infty)^2$ the normal cone is $N_K(x) = \{p : p_i \le 0 \text{ if } x_i = 0,\ p_i = 0 \text{ if } x_i > 0\}$, constant on each relative interior $\mathrm{relint}(F)$. For $x \in F_{\mathrm{int}}$ $(x_1, x_2 > 0)$: $N_K(x) = \{0\}$, giving L_{int}, 2-dimensional Lagrangian. For $x \in \mathrm{relint}(F_1)$ $(x_1 = 0 < x_2)$: $p = (-a, 0)$; $a > 0$ gives L_1 (Lagrangian, 2-dimensional) and $a = 0$ gives the ray R_1^+ (isotropic, 1-dimensional); similarly R_2^+ on F_2. At the vertex $x = 0$: $p = (-a,-b)$ with $a, b \ge 0$;

$a, b > 0$ gives L_0 (Lagrangian); $a > 0 = b$ gives R_1^-; $b > 0 = a$ gives R_2^-; $a = b = 0$ gives L_{00}. The nine pieces are pairwise disjoint (the sign patterns of (x, p) are mutually exclusive) and their union is all of ν_K^*; each is smooth, being a product of relative interiors. □

The projection cells $Q_F = \mathrm{relint}(F) + N_K(F)$ are the four closed quadrants; with $\tilde{Q}_F = \frac{1}{\sqrt{2}} Q_F$ the open cells $\tilde{Q}_F^\circ$ are the four open quadrants, pairwise disjoint.

Proposition 10.11 (Piecewise-quadratic S_K)**.** *For $K = [0, \infty)^2$ (a cone), $S_K(q) = \frac{1}{2}\|q\|^2 - \mathrm{dist}(q, K)^2$, i.e.*

$$S_K(q_1, q_2) = \begin{cases} \frac{q_1^2 + q_2^2}{2}, & q_1 > 0,\ q_2 > 0, \\ \frac{q_2^2 - q_1^2}{2}, & q_1 < 0,\ q_2 > 0, \\ \frac{q_1^2 - q_2^2}{2}, & q_1 > 0,\ q_2 < 0, \\ -\frac{q_1^2 + q_2^2}{2}, & q_1 < 0,\ q_2 < 0, \end{cases} \tag{7}$$

extended continuously to the axes. S_K is $C^{1,1}$ and C^1 across each facet: e.g. across $\{q_1 = 0\}$, $q_2 > 0$, $\partial_1 S \to 0$ from both sides while $\partial_{11}^2 S$ jumps $+1 \to -1$. The Hessian sign pattern $\mathrm{diag}(+1, +1)$, $\mathrm{diag}(-1, +1)$, $\mathrm{diag}(+1, -1)$, $\mathrm{diag}(-1, -1)$ *encodes the face: $+1$ on face-tangent (free) directions, -1 on normal (constrained) directions — the stratified Morse index data of Definition 2.2.*

Proof. For the cone $K = [0, \infty)^2$, $\mathrm{dist}(q, K)^2 = (q_1^-)^2 + (q_2^-)^2$ with $q_i^- = \max(-q_i, 0)$, so $S_K(q) = \frac{1}{2}(q_1^2 + q_2^2) - (q_1^-)^2 - (q_2^-)^2$, which is exactly (7) on the four open quadrants and extends continuously to the axes (all four formulas agree there). Across $\{q_1 = 0\}$, $q_2 > 0$: $\partial_1 S = q_1 \to 0$ from $q_1 > 0$ and $\partial_1 S = -q_1 \to 0$ from $q_1 < 0$, so S_K is C^1; similarly across $\{q_2 = 0\}$. The gradient ∇S_K is piecewise linear, hence globally Lipschitz: S_K is $C^{1,1}$. The Hessian is $\mathrm{diag}(\pm 1, \pm 1)$ on each quadrant, $+1$ in the directions tangent to the face containing the projection (free directions) and -1 in the normal directions (constrained directions) — the sign data of Definition 2.2. □

Proposition 10.12 (Per-stratum generation)**.** *Let $R(x, p) = (x, -p)$ and $C(x, p) = \left(\frac{x-p}{\sqrt{2}}, \frac{x+p}{\sqrt{2}}\right)$. For each face F,*

$$C(R(L_F)) = \mathrm{gph}(dS_F) \quad \text{over } \tilde{Q}_F^\circ,$$

where $S_F = S_K|_{\tilde{Q}_F^\circ}$ is the quadratic of (7), and the map is a diffeomorphism with explicit inverse. E.g. for F_1: $R(0, x_2, -a, 0) = (0, x_2, a, 0)$ and

$$C(0, x_2, a, 0) = \Big(\underbrace{-\tfrac{a}{\sqrt{2}}}_{q_1<0}, \underbrace{\tfrac{x_2}{\sqrt{2}}}_{q_2>0}\,;\, \underbrace{\tfrac{a}{\sqrt{2}}}_{-q_1}, \underbrace{\tfrac{x_2}{\sqrt{2}}}_{q_2}\Big),$$

so $(q, r) = (q_1, q_2, -q_1, q_2) = (q, dS_1(q))$ with $a = -\sqrt{2}\, q_1$, $x_2 = \sqrt{2}\, q_2$.

Proof. Write $C(x, p) = ((x - p)/\sqrt{2}, (x + p)/\sqrt{2}) =: (q, r)$. For $L_{\text{int}} = \{(x_1, x_2, 0, 0)\}$: R is the identity and C gives (q, q) with $q = (x_1, x_2)/\sqrt{2}$; $S_{\text{int}}(q) = \frac{1}{2}\|q\|^2$ has $dS_{\text{int}}(q) = q$. For $L_1 = \{(0, x_2, -a, 0)\}$, $a > 0$: R gives $(0, x_2, a, 0)$ and C gives $q = (-a/\sqrt{2}, x_2/\sqrt{2})$ ($q_1 < 0 < q_2$), $r = (a/\sqrt{2}, x_2/\sqrt{2}) = (-q_1, q_2) = dS_1(q)$ since $S_1(q) = (q_2^2 - q_1^2)/2$. The cases L_2 and L_0 are identical: $L_2 = \{(x_1, 0, 0, -b)\}$ gives $q = (x_1/\sqrt{2}, -b/\sqrt{2})$, $r = (q_1, -q_2) = dS_2(q)$; $L_0 = \{(0, 0, -a, -b)\}$ gives $q = (-a/\sqrt{2}, -b/\sqrt{2})$, $r = (-q_1, -q_2) = dS_0(q)$ with $S_0 = -(q_1^2 + q_2^2)/2$. In each case $(a, x_i) \leftrightarrow q$ is a linear isomorphism, so $C \circ R$ is a diffeomorphism onto its image, the graph of dS_F over the open projected cell $\tilde{Q}_F^\circ$. The rays $R_i^\pm$ map to the shared boundary hyperplanes $\{q_i = 0\}$, where the C^1 matching of Proposition 10.11 applies. □

Corollary 10.13 (Gluing at the orthant corner)**.** *The closed Lagrangian $C(R(\nu_K^*))$ for $K = [0, \infty)^2$ is the union of the four graphs $\mathrm{gph}(dS_F)$ over the closed projected cells, glued $C^{1,1}$ (not C^2) along the facet hyperplanes $\{q_i = 0\}$; each facet graph is covered twice by the two adjacent closures, in agreement (over $\{q_1 = 0,\, q_2 > 0\}$ both give $r = (0, q_2)$); all four closures meet at the origin. No fiber variables occur in any stratum.*

Proof. This is the assembly of Propositions 10.10, 10.11, and 10.12: the four strata give four quadratic graphs, and the explicit formulas show the two adjacent closures agree on each shared facet while the second derivatives jump, giving $C^{1,1}$ regularity. □

Remark 10.14 (Relation to the 1-dimensional case). Restricting (7) to either axis recovers $S_K(q) = \frac{q^2}{2}\operatorname{sgn}(q)$ for $K = [0, \infty)$. The quadrant decomposition here is also the target of the branched cover $\pi(z_1, z_2) = (z_1^2, z_2^2)$, which sends each $R(L_F)$ homeomorphically onto its quadrant.

Theorem 10.15 (Spectral invariants equal minimax: unconstrained; classical)**.** *For $K = \mathbb{R}^n$, the Lagrangian spectral invariants of the pair $(\mathrm{gph}(d\varphi)$, zero section — the action levels $c(\alpha) \in \mathbb{R}$ at which a homology class α appears in the action-filtered Floer complex — equal the minimax critical values of φ over the corresponding homology classes [42].*

Proof sketch. Viterbo's spectral invariants defined via generating functions quadratic at infinity coincide with the classical minimax values; see [42]. □

Proposition 10.16 (Finite-model spectral invariants equal classical minimax)**.** *Assume* (H1)–(H3)*:* (H1) *K compact convex polyhedron;* (H2) *every KKT point intrinsically nondegenerate (strict complementarity $-\nabla\varphi(x^*) \in \operatorname{ri} N_K(x^*)$ plus nonsingular face Hessian);* (H3) *φ smooth with bounded Hessian. Let (C_*, ∂) be the finite Morse complex of KKT generators with filtration $F_cC_* = \bigoplus_{\varphi(x^*)\le c} \mathbf{F}_2 \cdot x^*$, built from the ambient negative-gradient trajectories of the finite-dimensional (smoothed) model [15]. For $\alpha \in H_*(K;\mathbf{F}_2)$ put*

$$c_{\mathrm{KKT}}(\alpha) = \inf\{c : \alpha \in \operatorname{im}(H_*(F_cC_*) \to H_*(C_*))\}.$$

Then:

(i) *$H_*(F_cC_*) \cong H_*(K_c;\mathbf{F}_2)$ naturally, commuting with inclusions, where $K_c = \{\varphi \le c\} \cap K$;*

(ii) *$c_{\mathrm{KKT}}(\alpha)$ equals the classical sublevel minimax $c(\alpha) = \inf\{c : \alpha \in \operatorname{im}(H_*(K_c) \to H_*(K))\}$ and is a KKT critical value;*

(iii) *$c_{\mathrm{KKT}}(\alpha)$ and, more generally, all finite persistence intervals of the persistence module $V_c = H_*(F_cC_*)$ are computable from the filtered KKT complex and independent of the presentation $K = \{Ax \le b\}$: from grid data of mesh h a finite algorithm produces a barcode D_h with bottleneck distance $d_B(D_h, D(V)) \le Lh \to 0$ to the true barcode (grid-bridge convergence).*

Proof sketch. The commuting isomorphisms $H_*(F_cC_*) \cong H_*(K_c)$ are the filtered form of the finite-dimensional KKT complex [15]; a finite filtered complex changes only at its generator values, so the spectral value of a nonzero final class is a generator value, and the same argument applies to the entire persistence module. □

Conjecture 4. *In the setting of Conjecture 2, grant that $HF_*(\operatorname{gph}(\nabla\varphi), R(\nu_K^*))$ exists with an action filtration $HF_*^{<c}$ such that the action of a KKT generator is $\varphi(x^*)$. For $\alpha \in HF_*$ put*

$$c^{\mathrm{Floer}}(\alpha) = \inf\{c : \alpha \in \operatorname{im}(HF_*^{<c} \to HF_*)\}.$$

Then, via the filtration-preserving PSS isomorphism of Conjecture 2,

$$c^{\mathrm{Floer}}(\alpha) = c^{\mathrm{Morse}}(\alpha) = c(\alpha),$$

the finite-model (classical) minimax value of Proposition 10.16.

Proposition 10.17 (Mountain-pass as a finite-bar death)**.** *For convex K (hence contractible) the only nonzero final class is the degree-zero class, whose spectral value is* $\min_K \varphi$*. A mountain-pass level is* not *the spectral value of a nonzero class in $H_*(K)$: it is the* death *of a finite degree-zero persistence interval — two sublevel components merging at a saddle.*

Example. $K = [-4, 4]$, $\varphi = \cos x$ has KKT points $-\pi, 0, \pi$ of values $-1, 1, -1$ and indices $0, 1, 0$; the persistence module has one infinite bar (born -1) and one finite bar $[-1, 1)$, whose death 1 is the mountain-pass level, while the final H_0 spectral value is $-1 \neq 1$ (cf. the finite-model analysis in [15]). □

The appropriate objects are relative classes or finite persistence intervals, not absolute spectral invariants.

Proposition 10.18 (The phase-space set; proved)**.** *Let T be maximal monotone and put $\Lambda_T = \mathrm{gph}(T) \subset T^*X$. Then Λ_T is closed, and $\Lambda_T \cap R(\nu_K^*)$ projects* surjectively *onto the solution set of $0 \in T(x) + N_K(x)$, with fiber over a solution x^* equal to $T(x^*) \cap (-N_K(x^*))$. In the smooth single-valued case the fiber is a singleton and the projection is bijective; in general it is not. In the smooth single-valued case Λ_T is Lagrangian if and only if DT is symmetric, i.e. $T = \nabla\varphi$ locally (Proposition 3.1); for general maximal monotone T it is a closed monotone set but not Lagrangian.*

Proof. Closedness is maximal monotonicity; the surjection unwinds the definitions exactly as in Proposition 4.1. The Lagrangian characterization is Proposition 3.1. □

Program 10.19 (The maximal monotone horizon)**.** Develop a Floer-type invariant for closed maximal monotone (not necessarily Lagrangian) sets $\Lambda_T \subset T^*X$ recovering Theorems 10.1–10.3, 10.15 and Conjectures 2–4 when $T = \nabla\varphi$.

11 From the Arnold conjecture to optimization

The purpose of this section is to make the Arnold-to-optimization reduction explicit in three steps, and to isolate exactly what the variational route still needs. Arnold conjectured the lower bound; Floer proved it by infinite-dimensional Morse theory. The key point is that the fixed-point form $\mathrm{Fix}(\phi) = \Delta \cap \mathrm{gph}(\phi)$ is already a Lagrangian intersection — the dictionary only has to recognize it.

We make the reduction explicit: the Arnold conjecture, translated through our dictionary, becomes a statement about constrained optimization.

Theorem 11.1 (Arnold; fixed-point form [33])**.** *Let ϕ be a nondegenerate Hamiltonian diffeomorphism of a closed symplectic manifold (M, ω). Then* $\#\mathrm{Fix}(\phi) \geq \sum_i \dim H_i(M; \mathbf{F}_2)$.

Conjecture 5 (Arnold for variational inequalities)**.** *Let $K = \{x \in \mathbb{R}^n : Ax \leq b\}$ be a compact convex polyhedron and $\varphi : \mathbb{R}^n \to \mathbb{R}$ smooth, satisfying (H1)–(H4) of Conjecture 6. Let $(C_*^{\mathrm{KKT}}, \partial)$ be the complex generated by the KKT points (Definition 2.1), $C_k^{\mathrm{KKT}} = \bigoplus_{\mathrm{ind}(x^*)=k} \mathbf{F}_2 \cdot x^*$ graded by the stratified Morse index (Definition 2.2), with*

$$\partial x^* \;=\; \sum_{\mathrm{ind}(y^*)=\mathrm{ind}(x^*)-1} \#\mathcal{M}(x^*, y^*)\, y^* \pmod 2$$

counting isolated stratified connecting orbits. Then $\partial^2 = 0$ and

$$H_*(C_*^{\mathrm{KKT}}, \partial) \;\cong\; HM_*(\varphi, K; \mathbf{F}_2),$$

the stratified Morse homology of $\varphi|_K$. In particular,

$$\#\{\textit{KKT points}\} \;\geq\; \sum_i \dim HM_i(\varphi, K; \mathbf{F}_2).$$

This is the homological content of Conjecture 6: parts (a)–(d) there are exactly the well-definedness, $\partial^2 = 0$, and invariance asserted here. For K convex (hence contractible) the numerical bound gives $\#\mathrm{KKT} \geq 1$ — mere existence. The substance is the homology isomorphism.

Step 1: fixed points are Lagrangian intersections. $\mathrm{Fix}(\phi) = \Delta \cap \mathrm{graph}(\phi)$ in $(M \times M, \omega \oplus -\omega)$, so Arnold is a Lagrangian intersection lower bound. The relative form is Arnold–Givental: for τ anti-symplectic with $L = \mathrm{Fix}(\tau)$, $\#(\phi(L) \cap L) \geq \dim H_*(L; \mathbf{F}_2)$, announced by Bai–Shelukhin–Wang–Xu [5] (August 2026, awaiting verification). Our involution $R(x, p) = (x, -p)$ is of this type, with fixed set the zero section.

Step 2: the classical reduction to optimization. Via generating functions [42], $\mathrm{graph}(\phi) = \Gamma_S$ and $\mathrm{Fix}(\phi) \leftrightarrow \mathrm{Crit}(S)$; Arnold becomes $\#\mathrm{crit}(S) \geq \sum b_i$, i.e. Morse/Lusternik–Schnirelmann theory. This is the classical "Arnold reduces to optimization" (see e.g. [33]).

Step 3: our reduction. For the pair of our program, $L_0 = \mathrm{gph}(\nabla\varphi)$ and $L_1 = R(\nu_K^*)$, Proposition 4.1 gives $L_0 \cap L_1 \leftrightarrow$ KKT points. An Arnold-type bound $\#(L_0 \cap L_1) \geq \mathrm{rank}\, HF_*$ therefore becomes a multiplicity bound on KKT points — a statement purely about constrained optimization, made explicit in Conjecture 5 above.

Remark 11.2 (A singular generalization, not a case, of Arnold–Givental). Our program is *not* a special case of Arnold–Givental: Bai et al. require L smooth, whereas our $R(\nu_K^*)$ is singular for inequality constraints. It is a *singular* generalization, independent of their theorem — and conversely their generic ϕ is not our structured pair, so neither implies the other. The smooth sibling is the equality-constrained problem: for an orthogonal involution σ on $\mathbb{R}^n$ with fixed submanifold S, the anti-symplectic lift $\tau(x,p) = (\sigma(x), -(d\sigma_x^{-1})^* p)$ has $\mathrm{Fix}(\tau) = \nu_S^*$, the smooth conormal, corresponding to $x^* \in S$, $-\nabla\psi(x^*) \in (T_{x^*}S)^\perp$ (Lagrange multipliers). Two obstructions block a direct transfer of the Arnold–Givental bound to Lagrange counts: (1) non-compactness — ν_S^* has non-compact fibers, so a wrapped/non-compact Arnold–Givental would be needed [program]; (2) the fiber-preserving dead end of Lemma 11.3 below.

Lemma 11.3 (Fiber-preserving transfer is a dead end). *For $H(x,p) = \psi(x)$ the Hamiltonian flow ϕ_t preserves cotangent fibers; hence $\phi_t(\nu_S^*) \cap \nu_S^*$ is positive-dimensional over each Lagrange point. The naive fiber-preserving transfer of the Arnold–Givental bound to Lagrange counts is therefore badly non-transverse and incompatible with the generic-ϕ hypothesis.*

Proof. Hamilton's equations for $H(x,p) = \psi(x)$ give $\dot{x} = \partial H/\partial p = 0$ and $\dot{p} = -\partial H/\partial x = -\nabla\psi(x)$, so $\phi_t(x,p) = (x,\, p - t\nabla\psi(x))$: each fiber $\{x\} \times T_x^*\mathbb{R}^n$ is invariant. At a Lagrange point $x^* \in S$ with $-\nabla\psi(x^*) \in (T_{x^*}S)^\perp$, both p and $\nabla\psi(x^*)$ lie in $(T_{x^*}S)^\perp$ for $(x^*,p) \in \nu_S^*$, hence the whole line $\{(x^*,\, p - t\nabla\psi(x^*)) : t \in \mathbb{R}\}$ lies in $\phi_t(\nu_S^*) \cap \nu_S^*$. □

12 The variational counterparts of the missing theories

The purpose of this section is to give, for each missing symplectic theory, its variational-inequality counterpart — and to show that in several cases the variational side is already computable. The asymmetry noted in the introduction reverses here: the sixty-year variational theory leads, and the symplectic side follows. Our approach is closest to the finite-dimensional reduction philosophy of Viterbo [42], and can be compared with the sheaf route via Nadler–Zaslow [35].

The program above is stated symplectically, but each missing theory has a variational-inequality counterpart — and in several cases the variational side is the more developed one, so that the proof may go by translating *back* from variational inequalities to symplectic geometry. We state

the counterparts as full conjectures, self-contained in variational language, for direct attack.

Conjecture 6 (Conjecture 5′)**.** *Counterpart of Conjecture 2. Let* $K = \{x \in \mathbb{R}^n : Ax \le b\}$ *be a convex polyhedron and* $\varphi : \mathbb{R}^n \to \mathbb{R}$ *smooth. Assume:*

(H1) every KKT point (Definition 2.1) is nondegenerate in the sense of Definition 2.2, with stratified Morse index $\mathrm{ind}(x^*)$*;*

(H2) On faces F *with* $\dim F \le 2$ *(hence ambient* $n \le 3$*),* (G^+)*: nondegenerate folds,* $\mu_i^F \pitchfork 0$*, and* $\alpha_i^F(b) \neq 0$ *at KKT points — which implies* (Z_{conn}) *. On faces of dimension* ≥ 3*,* (Z_{conn}) *is assumed explicitly: every stratified connecting orbit undergoes finitely many face transitions, with no Zeno accumulation of switch points.*

(H3) (NG-global): no connecting orbit, and no limit of connecting orbits, meets a deactivation locus tangentially without crossing it;

(H4) g *belongs to a fixed residual set of Riemannian metrics in the* C^k *Whitney topology (*k *large enough for the parametric transversality machine), for which every* $\mathcal{M}(a,b)$ *is stratified-transverse.*

A stratified anti-gradient trajectory *is an absolutely continuous* $x(t) \in K$ *with* $\dot{x} = -\operatorname{grad}_g(\varphi|_F)$ *on each relatively open face* F*, whose face transitions occur only at transverse crossings of deactivation/activation loci (nonzero crossing speed of the face multiplier field* $\tilde{\mu}_i$*). Then:*

(a) if $\mathrm{ind}(a) - \mathrm{ind}(b) = 1$*, the space* $\mathcal{M}(a,b)$ *of unparametrized trajectories from* a *to* b *is finite;*

(b) $\partial a = \sum_{\mathrm{ind}(b)=\mathrm{ind}(a)-1} \#\mathcal{M}(a,b)\, b \pmod 2$ *satisfies* $\partial^2 = 0$*;*

(c) $HM_*(\varphi, K; g) = H_*(C_*, \partial)$*,* $C_k = \bigoplus_{\mathrm{ind}=k} \mathbb{Z}_2 \cdot x^*$*, is independent of* g *and invariant under homotopies* φ_t *through nondegenerate data;*

(d) $HM_*(\varphi, K) \cong H_*(K; \mathbb{Z}_2)$*.*

Remark 12.1. Parts (a) and (b) are proved under Hypotheses 12.5, 12.6, and 12.7 (Theorems 12.8 and 12.9); part (c) is partially proved.

The results on Conjecture 5 proved here are Theorems 12.8 and 12.9 below.

Lemma 12.2 (No fake equilibria on larger faces)**.** *Let* $b \in \mathrm{relint}(F_b)$ *be KKT with strict complementarity (*$\lambda_i > 0$ *for every active constraint) and* $F_a \supsetneq F_b$*, under the usual nondegeneracy (LICQ). Then* b *is* not *an equilibrium of the smooth flow on* F_a*.*

Proof. Write $-\nabla\varphi(b) = \sum_{i\in I_b} \lambda_i \nabla g_i$. The part over I_a lies in the normal space $N_{F_a}(b)$ and projects to zero on T_{F_a}; the remainder $w = \sum_{i\in I_b\setminus I_a} \lambda_i \nabla g_i$ is a strictly positive combination of gradients outside I_a, hence $w \notin \operatorname{span}\{\nabla g_i : i \in I_a\} = N_{F_a}(b)$ by linear independence, so $P_{T_{F_a}}(-\nabla\varphi(b)) \neq 0$. Downward face-transitions therefore converge to the boundary cleanly, with nonzero normal velocity — no extra genericity hypothesis is needed. □

Remark 12.3 (Warning: naive genericity does not control the 2-jet). On $[0,1]^2$, $\varphi(x,y) = ((x-\frac{1}{2})+y)^2 - y$ has a KKT point $b = (\frac{1}{2}, 0)$ that is nondegenerate with strict complementarity, yet $\nabla^2\varphi(b) = \left(\begin{smallmatrix} 2 & 2 \\ 2 & 2 \end{smallmatrix}\right)$ is singular. Any argument needing the full Hessian must state its jet hypothesis explicitly. [Computed October 2026; code in agent-notes.]

Proposition 12.4 (Lift-off criterion)**.** *For a face F with active constraint i, let $\tilde{\mu}_i(x) = [(AA^\top)^{-1}A(-\nabla\varphi(x))]_i$ be the face multiplier (A the outward-normal matrix; in words, the i-th component of the KKT multiplier field extended off the face). The formula via $(AA^\top)^{-1}$ requires the active rows to be independent; intrinsically, $\tilde{\mu}$ is the component of $-\nabla\varphi(x)$ in the normal cone $N_F(x)$, well-defined without LICQ, and every invocation below is understood in this intrinsic sense (or by restriction to a LICQ sub-presentation). If $\tilde{\mu}_i$ changes sign $+ \to -$ along an orbit while the other multipliers stay positive, the orbit leaves F for the face with active set $I \setminus \{i\}$, tangent to F at the switch point. The generic case is a transverse crossing, open in φ.*

Proof. Let $I = I(F)$ be the active set on F and A the matrix of outward normals. The face multiplier field $\tilde{\mu}(x) = (AA^\top)^{-1}A(-\nabla\varphi(x))$ extends the KKT multipliers off F; intrinsically it is the component of $-\nabla\varphi(x)$ in the normal cone $N_F(x)$, hence well defined without LICQ. The deactivation locus for constraint i is $D_{F,i} = \{\tilde{\mu}_i = 0\}$: $\tilde{\mu}_i > 0$ means constraint i presses the orbit onto F, $\tilde{\mu}_i < 0$ means it releases. If $\tilde{\mu}_i$ changes sign $+ \to -$ along an orbit γ at $p = \gamma(0)$ while the other multipliers stay positive, then p is a lift-off point: before p the orbit lies on F with i active, after p constraint i is inactive, so the orbit continues in the face with active set $I \setminus \{i\}$. At p itself $\tilde{\mu}_i(p) = 0$, i.e. the normal velocity component vanishes, so the orbit is tangent to F at the switch point. The generic case is a transverse crossing $d\tilde{\mu}_i[v_F](p) \neq 0$ (equivalently < 0 for a $+ \to -$ change), which is open in φ; degenerate lift-off (vanishing derivative with sign change) is codimension 3 in jet space, hence avoided by a residual set of φ. □

The next two theorems are proved under three explicit analytic hypotheses on the moduli spaces of stratified connecting orbits. We state them here as hypotheses; proving them is the remaining analytic work.

Hypothesis 12.5 (Compactness)**.** Under (H1) plus (Z_{conn}), the closure of $\mathcal{M}(a,b)$ in C^0_{loc} modulo time shift is compact, and its boundary consists only of broken trajectories through KKT points.

Hypothesis 12.6 (Transversality)**.** Under (H1), (Z_{conn}), and g residual in the C^k Whitney topology, every $\mathcal{M}(a,b)$ is stratified-transverse, of expected dimension $\mathrm{ind}(a) - \mathrm{ind}(b) - 1 - c(w)$.

Hypothesis 12.7 (Gluing)**.** Every broken pair is the endpoint of a unique 1-parameter gluing family.

Theorem 12.8 (Finiteness)**.** *Assume (H1)–(H4) of Conjecture* 5′ *and Hypotheses 12.5 and 12.6. If* $\mathrm{ind}(a) - \mathrm{ind}(b) = 1$*, the space* $\mathcal{M}(a,b)$ *of unparametrized stratified connecting orbits from* a *to* b *is finite.*

Proof. Hypothesis 12.6 makes $\mathcal{M}(a,b)$ a smooth 0-dimensional manifold, hence discrete: the main stratum has dimension $\Delta - 1 = 0$ (with $\Delta = \mathrm{ind}(a) - \mathrm{ind}(b)$), and degenerate strata have negative dimension, hence are empty. For $\Delta = 1$ no broken trajectory can appear in the compact closure of Hypothesis 12.5: a broken orbit $a \to c_1 \to \cdots \to c_k \to b$ ($k \geq 1$) drops the index by at least 1 on each leg, for a total drop $\geq k + 1 \geq 2$, against $\Delta = 1$ (using strict decrease of φ along nonconstant orbits). Thus $\mathcal{M}(a,b)$ equals its compact closure, hence is compact. Discrete and compact gives finite.

□

Theorem 12.9 ($\partial^2 = 0$)**.** *Assume (H1)–(H4) of Conjecture* 5′ *and Hypotheses 12.5 and 12.7. With* ∂ *defined by mod-*2 *counting as in Conjecture* 5′*(b),* $\partial^2 = 0$*.*

Proof. For $\mathrm{ind}(a) - \mathrm{ind}(c) = 2$, Hypothesis 12.5 compactifies $\mathcal{M}(a,c)$ to a compact 1-manifold with boundary, whose boundary points are exactly the once-broken trajectories $a \to b \to c$ with $\mathrm{ind}(b) = \mathrm{ind}(a) - 1$: multiple breaks are ruled out by the index-drop count, and breaks can only occur at KKT points. Hypothesis 12.7 puts each broken pair at the end of a unique 1-parameter gluing family. The boundary of a compact 1-manifold has even cardinality, so the mod-2 count of broken pairs cancels in pairs, which is exactly $\partial^2 = 0$.

□

Theorem 12.10 (Two-dimensional case)**.** *On a compact convex polygon* $K \subset \mathbb{R}^2$*, with* φ *generic (nondegenerate KKT points with strict complementarity, transverse hitting, Morse–Smale),* $\partial_1 \partial_2 = 0 \pmod 2$ *even when* ∂_2 *counts hitting orbits. The isolated hitting directions cut the circle of*

downward directions from each index-2 point into sectors; each sector is counted exactly twice around the circle. The argument is genuinely two-dimensional: in 3D isolated hitting points do not separate S^2, and index-2 saddles on faces have two-dimensional unstable manifolds.

Proof. Let P_2 be an index-2 KKT point. Consider the circle S^1 of downward directions from P_2 and the map $W(\theta)$ sending a direction θ to the ω-limit of the projected orbit it launches. Under the genericity hypotheses, W is locally constant: it can only jump at the isolated hitting directions θ_i (transverse hitting), because a vertex-hit causes no jump (index-0 vertices are sinks) and nonzero normal velocity forces finite-time landing, precluding any other jump mechanism. At a hitting direction θ_i with edge-saddle q_i (whose unstable manifold is 1-dimensional), the two sides of θ_i give the two branches of $W^u(q_i)$, so $W(\theta_i^-) + W(\theta_i^+) = \partial_1(q_i)$, the two index-0 endpoints. Ordering $\theta_1 < \cdots < \theta_m$ around the circle with $W \equiv c_j$ on the j-th sector gives $\sum_i \partial_1(q_i) = \sum_j (c_{j-1} + c_j) = 2\sum_j c_j = 0 \pmod 2$: every sector is counted exactly twice. Hence $\partial_1\partial_2(P_2) = 0 \pmod 2$ with hitting orbits counted. The argument is genuinely two-dimensional: it uses the cyclic order of S^1. In 3D the direction space is S^2 with no cyclic order forcing the pairing, and index-2 face saddles have 2-dimensional unstable manifolds with arbitrary connection counts, so $\partial_2\partial_3 \neq 0$ can occur stably. □

Theorem 12.11 (Hitting orbits break $\partial^2 = 0$ in dimension ≥ 3)**.** *In dimension ≥ 3, counting hitting orbits in the top differential ∂_3 breaks $\partial^2 = 0$: naive barrier-smoothing does not repair the 3D top layer (it is not well-defined, depending on the choice of barrier), and the question "to count or not to count" is ill-posed in the naive projected-flow framework. Hitting orbits are therefore not counted in ∂_3; they remain a known open obstruction, and the correct definition of the limit complex is open.*

Proof. The naive barrier-smoothing was refuted in dimension 3: a smoothed ε-complex still breaks, and two different barriers give two different answers, so the count is not well-defined. In dimension 2 the full theory closes (Theorem 12.10), where no hitting phenomenon occurs. Hence in dimension ≥ 3 the top differential is defined without hitting-orbit contributions, and the obstruction is recorded as open. □

Remark 12.12 (No-grazing as an explicit hypothesis). The stable-set identity is stated under an explicit no-grazing hypothesis (NG): no connecting orbit meets a deactivation locus tangentially without crossing it. (NG) cannot be proved from the other hypotheses, but it is a legitimate genericity-type hypothesis: the grazing set has measure zero, and genuine fold grazing exists, so the hypothesis is needed, not extra caution. Correction:

at a boundary point $b \in \partial F_a$ the identity is applied with the lower face containing b in its relative interior, not with F_a itself.

Proposition 12.13 (The $L^3 = 0$ case of 3D genericity)**.** *The last open case of the 3D face genericity (P2) holds on* $\{(L^3, (v_0)_3) \neq (0,0)\}$*: the* $\{L^3 \neq 0\}$ *part by three channels and the* $\{L^3 = 0, (v_0)_3 \neq 0\}$ *part by the Q-channel. The master identity* $d(\det M) = (v_0)_3 \cdot (\cdots)$ *was checked independently twice. The remaining exception* $\{L^3 = 0, (v_0)_3 = 0\}$ *is empty generically:* 0 *is a regular value, so the bad set has codimension* 6 *and a generic family contains no bad point. The paper does not use (P2) in any theorem.*

Proof. Let $\Phi = (\alpha, L_v\alpha, L_v^2\alpha)$ on a face, $\Sigma = \{\Phi = 0\}$, and at a rank-2 point put $M = D_x\Phi$, $u_0 = \ker M$, v_0 the left kernel, $L^3 = L_v^3\alpha$, and $V = (\Phi, \det M)$. For jet variations with $d\Phi = 0$, Jacobi's formula gives $d(\det M) = \operatorname{tr}(\operatorname{adj}(M)\, dM)$. On $\{L^3 \neq 0\}$ the fold direction u_0 is not parallel to v, and the symmetric-tensor lemma makes the H- and T-functionals independent of the constraints; transversality $d(\det M) \neq 0$ with $d\Phi = 0$ is then achieved by three channels: (Q) if $(v_0)_3 \neq 0$; (T) if $(v_0)_3 = 0 \neq (v_0)_2$, via $T \in \mathrm{Sym}^3$ with $\ell(T) = 0$, $T(a, v, u_0) \neq 0$; (H,T) if $(v_0)_3 = (v_0)_2 = 0$, via $H \in \mathrm{Sym}^2$ with $H(a, v) = 0 \neq H(a, u_0)$. On $\{L^3 = 0, (v_0)_3 \neq 0\}$ the Q-channel alone applies: a pure 4-jet bump gives $d(\det M) = (v_0)_3\, \ell(Q(v)) \neq 0$ with $d\Phi = 0$ automatically — the master identity $d(\det M) = (v_0)_3 \cdot (\cdots)$, checked independently twice. Hence V is transverse to 0 at every rank-2 point with $(L^3, (v_0)_3) \neq (0,0)$, and parametric transversality gives the claim for a residual set of families. The remaining bad set $\{L^3 = 0, (v_0)_3 = 0\}$ is cut out by six independent jet equations, hence codimension 6; a 1-parameter family has 3-dimensional domain, so generic families contain no bad point. (P2) is not used in any theorem of the paper. □

Lemma 12.14 (Non-redundant description gives LICQ)**.** *Hypothesis (K) gains one free condition: write the polyhedron with a* non-redundant description $K = \{Ax \leq b\}$*, meaning no constraint can be dropped without changing* K*. Every polyhedron has such a description, so no polyhedron is excluded. The gain is automatic LICQ: on a non-redundant description, the active gradients are linearly independent at every face of dimension* ≥ 1*. Hence the deactivation lemma no longer needs LICQ as a separate hypothesis, and non-simple vertices are kept.*

Proof. Work in $\mathbb{R}^3$, the setting where the lemma is used. Every polyhedron admits an irredundant H-description: drop redundant inequalities one by one; the process terminates with K unchanged, so no polyhedron is excluded. Let F have $\dim F \geq 1$ and $p \in \operatorname{relint}(F)$. If F is a facet

(dim 2), exactly one constraint is active with nonzero normal. If F is an edge (dim 1), the tangent cone is $T_K(p) = L \oplus C$ with L the edge line and C a pointed 2-dimensional cone in the transverse plane (pointed because the normal cone is 2-dimensional); a pointed 2D cone has exactly two extreme rays, so exactly two facets contain the edge, and the normals of two distinct planes through a line are linearly independent. Vertices may be non-simple; LICQ can only fail there. Thus with irredundant H, LICQ holds automatically on every face of dimension ≥ 1, and the deactivation lemma needs no separate LICQ hypothesis. □

Theorem 12.15 (Strong P-A2 in 3D, per-pair, mod 2)**.** *Let $K \subset \mathbb{R}^3$ be a compact convex polyhedron and φ smooth on a neighborhood of K. The KKT points are the equilibria of the projected gradient flow $\dot{x} = \Pi_{T_xK}(-\nabla\varphi(x))$, and* $\mathrm{ind}(p) := \dim W^u(p)$*. Assume:*

(K) K is compact, with a non-redundant description (Lemma 12.14);

(N1) every KKT point is nondegenerate: the reduced Hessian on the point's own face is invertible; in particular the critical set Crit *is finite;*

(N2) strict complementarity: every KKT multiplier is strictly positive;

(T) the explicit genericity package: (T-Ψ) the 3-jet section $\Psi_{\varphi,F,i}$ meets $\{0\}$ transversely on $\{v_F \neq 0\}$ for every proper face F and every constraint i that can be released from F; (T1) Kupka–Smale transversality inside each smooth face; (NG) no-grazing for the stable trace (Remark 12.12); (T-iiiB) the endpoint condition at a lower face, including injectivity of the endpoint map; (T-B-pair) the pair map (α_i, α_j) on every edge meets $\{0\}$ transversely; (T-corner) no connecting orbit passes through a vertex other than its two endpoints; (T-val) the values $\varphi(c)$, $c \in$ Crit, are pairwise distinct; and (T-end) the virtual multipliers at a KKT point do not vanish on any face through it: $\alpha_i^F(b) \neq 0$ for every $F \ni b$ and every $i \in I(F)$. Of these, (T-Ψ), (T1), (T-val) and (T-end) hold on residual sets of φ (for (T-Ψ) see Proposition 12.13); (T-B-pair) is expected to by the same 1-jet argument; (NG), (T-iiiB) and (T-corner) are imposed as explicit hypotheses;

(Z_{conn}) no-Zeno, in closure form: for every pair of distinct KKT points a, b, every connecting orbit in $\mathcal{M}(a, b)$ undergoes finitely many face switches, and the number of face switches is bounded uniformly over $\mathcal{M}(a, b)$; equivalently, the closure of $\mathcal{M}(a, b)$ in the topology of uniform convergence on compact sets contains no orbit with a Zeno episode.

The cell-side count is defined as follows. For $\mathrm{ind}(a) = \lambda \geq 1$ *let* $S_a := W^u_{\mathrm{loc}}(a) \cap \{\varphi = \varphi(a) - \delta\}$ *be the attaching sphere* ($\delta > 0$ *small), choose* ℓ *with* $\max\{\varphi(c) : c \in \mathrm{Crit},\ \varphi(c) < \varphi(a)\} < \ell < \varphi(a)$ *a regular value on every stratum, put* $L := \{\varphi = \ell\}$*, let* $h : S_a \to L$ *be the level-set hitting map (the first point at which the forward orbit through a point of* S_a *meets* L*), and put* $\Sigma_b := W^s(b) \cap L$*. Set*

$$n_{\mathrm{cell}}(a, b) := \#\big(h(S_a) \cap \Sigma_b\big) \pmod 2,$$

equivalently the mod-2 degree of the composite $S_a \xrightarrow{h} L \to S^{\lambda-1}$ *obtained from* h *by the Thom collapse of a tubular neighborhood of* Σ_b *in* L*. With* $n_{\mathrm{orbit}}(a, b) := \#\mathcal{M}(a, b)/\mathbb{R} \pmod 2$*, where hitting orbits are counted as connecting orbits (an orbit that reaches its target in finite time with nonzero velocity transverse to the target face is a connecting orbit), the conclusion is*

$$\mathrm{ind}(a) - \mathrm{ind}(b) = 1 \implies n_{\mathrm{orbit}}(a, b) = n_{\mathrm{cell}}(a, b) \pmod 2$$

for every pair of distinct KKT points a, b*. Hence* $\partial_{\mathrm{orbit}} = \partial_{\mathrm{cell}}$ *as operators on* $C_* = \bigoplus_p \mathbf{F}_2 \cdot e_p$*, per pair, and the* φ*-filtration is compatible on both sides.*

Proof. We argue per pair (a, b) with $\mathrm{ind}(a) - \mathrm{ind}(b) = 1$ and write $\lambda = \mathrm{ind}(a)$. The proof has six steps.

Step (i): the local unstable manifold and the attaching sphere. Near a KKT point a satisfying (N2), write $w^* = -\nabla\varphi(a) = \sum_{i \in I(a)} \lambda_i a_i$ with all $\lambda_i > 0$; hence $w^* \in \mathrm{ri}(N_K(a))$, the projection onto the normal cone is affine near w^*, and the projected field agrees on $\mathrm{relint}(F_a)$ near a with the smooth field $x \mapsto \Pi_{T_{F_a}}(-\nabla\varphi(x))$, whose derivative at a is minus the reduced Hessian. In adapted coordinates (y, z) at a, with $z_i = g_i(x) \leq 0$ the active constraints, one has $\dot{z}_i > 0$ when $z_i < 0$: the face F_a is locally forward invariant and normally attracting, so an orbit with α-limit a eventually lies in F_a. By (N1), a is a hyperbolic equilibrium of the smooth face field, and the classical unstable-manifold theorem gives a unique germ $W^u_{\mathrm{loc}}(a)$: a smooth embedded λ-disk in $\mathrm{relint}(F_a)$, tangent to $E^u(a)$ at a. Along every nonconstant orbit, $\frac{d}{dt}\varphi = -\|v\|^2 < 0$ almost everywhere, and $v(x) = 0$ if and only if x is a KKT point; hence φ is a strict Lyapunov function, there are no nonconstant periodic orbits, and by La Salle's invariance principle, with (N1) (KKT points are isolated, and Crit is finite by compactness), every forward orbit converges to a single KKT point. For $\delta > 0$ small, $S_a := W^u_{\mathrm{loc}}(a) \cap \{\varphi = \varphi(a) - \delta\} \cong S^{\lambda-1}$ is the attaching sphere; it is well defined up to isotopy. Only (K), (N1), (N2) are used in this step.

Step (ii): the cell count via the level-set hitting map. Since Crit is finite and its values are pairwise distinct by (T-val), choose ℓ with $\max\{\varphi(c) : c \in \mathrm{Crit},\ \varphi(c) < \varphi(a)\} < \ell < \varphi(a)$, regular on every stratum (possible by Sard's theorem); then $L = \{\varphi = \ell\}$ is compact and contains no KKT point. For $x \in S_a$, the function $t \mapsto \varphi(\Phi_t(x))$ is continuous and strictly decreasing, from a value $< \varphi(a)$ down to $\varphi(b_x) < \ell$, where b_x is the limit KKT point of Step (i); hence there is a unique hitting time $\tau(x) \geq 0$ with $\varphi(\Phi_{\tau(x)}(x)) = \ell$. The hitting time is continuous: if $\tau(x_n) \to \infty$ along a subsequence for $x_n \to x_0$, then $\varphi(\Phi_T(x_0)) \geq \ell$ for every T, contradicting $\varphi(\Phi_T(x_0)) \to \varphi(b_{x_0}) < \ell$; upper and lower semicontinuity then follow from strict decrease and joint continuity of the projected semiflow at finite times. Thus $h : S_a \to L$, $h(x) = \Phi_{\tau(x)}(x)$, is continuous. Under the transversality input of Step (iii), $\Sigma_b = W^s(b) \cap L$ is a smooth compact submanifold of L of codimension $\lambda - 1$; after a small perturbation h is transverse to Σ_b, and $h(S_a) \cap \Sigma_b$ is finite. The mod-2 intersection number is independent of ℓ, δ, the perturbation and the metric, and by Pontryagin–Thom it equals the mod-2 degree of the composite $S_a \xrightarrow{h} L \to S^{\lambda-1}$ given by the Thom collapse of a tubular neighborhood of Σ_b in L; this is the definition of $n_{\mathrm{cell}}(a, b)$ in the statement. It replaces the earlier definition of the cell count through a deformation retraction of the sublevel set onto its skeleton: that retraction has not been constructed, while the present definition is constructed here, and the filtration property is direct — $n_{\mathrm{cell}}(a, b) \neq 0$ implies that a connecting orbit $a \to b$ exists (Steps (iii)–(iv) below), hence $\varphi(b) < \varphi(a)$ by strict decrease. The case $\lambda = 1$ is included: $S_a = S^0$, and the count is the number of branches of $W^u(a)$ flowing to b, mod 2.

Step (iii): transversality and the dimension formula. Fix a connecting orbit γ and its finite face word $w(\gamma)$, the sequence of faces its legs lie in (finiteness on each orbit is part of ($\mathrm{Z}_{\mathrm{conn}}$)); nearby orbits have the same word, since transverse switches are open conditions. Two independent counts show that the switches contribute nothing to the dimension.

Incidence count. Parametrize candidate orbits with word w by $(x, t_1, \ldots, t_k)$, where $x \in S_a$ and the t_j are the switching times. Each type-(A) switch imposes one equation $\alpha_{i_j}(\gamma(t_j)) = 0$, whose derivative in t_j is $\dot{h}_j = d\alpha_{i_j}[v](p_j) \neq 0$ by the order-one transversality of Step (iv); the incidence Jacobian is block lower triangular with diagonal entries $\dot{h}_j \neq 0$, so each switch contributes $+1$ parameter and -1 equation: net 0.

Fredholm count. The linearization along γ is $L\xi = \dot{\xi} - A(t)\xi$ on the smooth legs, with transmission conditions $\xi(t_j^+) = S_j \xi(t_j^-)$ at the switches. At a type-(A) switch the velocity is continuous, so $S_j = I$; at a touch-down (type-(R)) switch, $S_j = P_{T_{F'}}$, the projection onto the tangent space of the

face entered; at a simultaneous switch the transmission is again the identity. In each case the transmission domain $W_j = \mathrm{graph}(S_j)$ has dimension $n = 3$ in $\mathbb{R}^6$, so by the index formula with transmission points each switch contributes $\dim W_j - n = 0$, and $\operatorname{ind} L = \dim E^u(A_-) - \dim E^u(A_+) = \operatorname{ind}(a) - \operatorname{ind}(b)$, independently of the face word. The time-translation field $\dot\gamma$ lies in the kernel, since $S_j\dot\gamma(t_j^-) = \dot\gamma(t_j^+)$ for every switch type. The moduli space is a manifold in this transmission frame; in naive coordinates it is only C^0 at orbits with a touch-down switch, which is enough for the mod-2 count.

Transversality at γ is surjectivity of L. On the last smooth leg it is supplied by (T1), the identification of the stable trace on a section with the stable manifold of the face flow being the content of (NG). At a lower-face endpoint $F_b \subsetneq F_a$, the arrival is clean for free from (N2): if the tail lies in a strictly larger face $F' \supsetneq F_b$, then $v_{F'}(b) \neq 0$ with nonzero component normal to F_b, so γ reaches b in finite time, transversely to F_b; the remaining finite-dimensional condition is (T-iiiB), that the endpoint map $E : U \subset S_a \to \Sigma$ to a section Σ placed before b is injective with local degree ± 1, whence γ is isolated and contributes exactly 1 mod 2. With $\operatorname{ind} L = \operatorname{ind}(a) - \operatorname{ind}(b)$, the quotient by time translation gives $\dim \mathcal{M}(a,b)/\mathbb{R} = \operatorname{ind}(a) - \operatorname{ind}(b) - 1$; for index difference 1 every connecting orbit is isolated.

Step (iv): only transverse type-(A) meetings occur. For a proper face F and a releasable constraint i, let α_i be the virtual multiplier on $\mathrm{relint}(F)$ and $\Psi_{\varphi,F,i} = (\alpha_i, L\alpha_i, L^2\alpha_i)$ with $L = L_{v_F}$. If a sign-changing lift-off at $p \in \mathrm{relint}(F)$ is degenerate — the function $h(t) = \alpha_i(\gamma(t))$ changes sign $+ \to -$ with $\dot h(p) = 0$ — then also $\ddot h(p) = 0$: a zero of order exactly 2 cannot change sign. Hence a degenerate sign-changing lift-off forces $\Psi_{\varphi,F,i}(p) = 0$. The three components of Ψ are independent conditions on the 3-jet of φ: bump perturbations in the 1-jet, 2-jet and 3-jet change exactly one component at a time (a triangular structure), so the degenerate locus has codimension exactly 3. By (T-Ψ) it is transverse, hence of dimension $\dim F - 3 \leq -1$ for every proper face (in $\mathbb{R}^3$, $\dim F \leq 2$): it is empty on $\{v_F \neq 0\}$. An interior switch point of a connecting orbit has $v_F(p) \neq 0$: otherwise p is an equilibrium of the smooth face field, the orbit through p is constant on F, or p is a KKT point — a breaking point, not a switch. Therefore every simple sign-changing lift-off in 3D is transverse of order one, and no cubic points occur. This is a genuinely three-dimensional phenomenon: on a 3-dimensional face (ambient dimension ≥ 4) the same count leaves isolated cubic points, which persist under perturbation. Grazing contacts ($\alpha_i = L\alpha_i = 0$, $L^2\alpha_i > 0$) are not switches: the face word does not change at them, and the smooth arguments pass through. Simultaneous switches are excluded by (T-B-pair), and passages

through a vertex other than the endpoints by (T-corner).

Step (v): the index enters only through the Fredholm index. No static identification of indices on different faces is used or needed. At its own face, the KKT index is $\dim E^u$ by definition: W^u at a strict-complementarity KKT point lies in its own face, and the normal directions are attracted. For nested faces the Schur formula $\mathrm{ind}_F(b) = \mathrm{ind}_{\mathrm{KKT}}(b) + \mathrm{ind}(S)$, with S the Schur complement of the reduced Hessian, shows that the correction $\mathrm{ind}(S)$ is independent data, and the static identity $\mathrm{ind}_{F_a}(b) = \mathrm{ind}_{\mathrm{KKT}}(b)$ is false in general. What the proof uses is only the conclusion of Step (iii): the Fredholm index of the linearized operator along any connecting orbit, with transmission conditions, equals $\mathrm{ind}_{\mathrm{KKT}}(a) - \mathrm{ind}_{\mathrm{KKT}}(b)$. Consequently a nonconstant broken piece $c \to c'$ with nonempty transverse moduli satisfies $\mathrm{ind}_{\mathrm{KKT}}(c) - \mathrm{ind}_{\mathrm{KKT}}(c') \geq 1$.

[One point is not proved. For a broken piece that enters the target face in finite time and then converges inside that face, the index formula holds verbatim: the hitting event is an interior transmission point whose saltation is a projection, of index contribution 0. For a piece that reaches its endpoint c' itself in finite time, the formula is proved for the extended operator L_{hit}, built with the stopping transmission condition $W_{\mathrm{stop}} = \mathbb{R}^n \times \{0\}$ at the hitting time; its kernel contains the extended time-translation field. But surjectivity of L_{hit} — condition (T-hit) — is not proved. It is expected to be a residual condition, by a bump argument on the last smooth leg, and it does not follow from (T-Ψ) or (T1). For index difference 1 the counting conclusion of this theorem is unaffected, since Step (iii) supplies the local degree directly; (T-hit) is used only here, to exclude intermediate hitting pieces of arbitrary index drop in the compactness argument of Step (vi). This is the single open point of the six-step chain.]

Step (vi): compactness and assembly. Let $[\gamma_k] \in \mathcal{M}(a,b)/\mathbb{R}$; normalize representatives by $\varphi(\gamma_k(0)) = \frac{\varphi(a)+\varphi(b)}{2}$, possible since φ decreases strictly and crosses the mid-level exactly once. The orbits are uniformly Lipschitz on the compact set K; by Arzelà–Ascoli a subsequence converges uniformly on compact sets to an orbit of the projected flow, and every rest point of the limit is a genuine KKT point (a rest point of the projected flow is a KKT point, by the Moreau decomposition). The limit is a broken trajectory through KKT points $a = c_0 \to c_1 \to \cdots \to c_m = b$, each piece converging to a single KKT point at each end by La Salle (Step (i)). By Step (v) each nonconstant piece drops the KKT index by at least 1, so $\mathrm{ind}(a) - \mathrm{ind}(b) \geq m$; for index difference 1, $m = 1$: there is no intermediate breaking point. It remains to exclude accumulation of switches inside a single piece. Switch points accumulating at a finite time

t^* have limits that are transverse type-(A) switches (which persist, with a uniform time gap between consecutive clean excursions from the same face–constraint pair), cubic points (excluded in Step (iv)), or grazing points. A grazing limit is not a switch: the face word of the limit orbit is obtained from the common word by deleting the slots that degenerate to grazing, so the limit is still a connecting orbit in $\mathcal{M}(a,b)$. Accumulation at an endpoint is excluded because the virtual multipliers are strictly positive near a strict-complementarity KKT point (and, by (T-end), nonzero on every face through it), so a connecting orbit has only finitely many switches near the endpoint. Every remaining mechanism requires infinitely many switches along the sequence, or a Zeno episode in the closure of $\mathcal{M}(a,b)$; both are excluded by ($\mathrm{Z}_{\mathrm{conn}}$) in the closure form stated above. Hence $\mathcal{M}(a,b)/\mathbb{R}$ is compact; with Step (iii) it is a compact 0-dimensional space, therefore finite, and $n_{\mathrm{orbit}}(a,b)$ is well defined. Finally, the bijection of Steps (ii)–(iii) identifies the regular preimages of the Thom collapse with the connecting orbits one-to-one, so $n_{\mathrm{cell}}(a,b) = \#\mathcal{M}(a,b)/\mathbb{R} \pmod 2 = n_{\mathrm{orbit}}(a,b)$. Summing over all b gives $\partial_{\mathrm{orbit}} = \partial_{\mathrm{cell}}$ on C_*, per pair. □

Conjecture 7. *Counterparts of Conjectures 3 and 4. Fix a compact convex polyhedron K and transverse Morse φ as in Conjecture 6. A* presentation *is any (A,b) with $K = \{Ax \le b\}$. Let (C_*, ∂) be the filtered KKT complex, $F_cC_* = \bigoplus_{\varphi(x^*)\le c} \mathbf{F}_2 \cdot x^*$, with persistence module $V_c = H_*(F_cC_*)$; in the finite (ambient, smoothed) model this is Proposition 10.16, and its identification with the face-by-face complex of Conjecture 6 is conditional on parts (a)–(c) there. For $\alpha \in H_*(K;\mathbf{F}_2)$ put*

$$c_{\mathrm{KKT}}(\alpha; A, b) = \inf\{c : \alpha \in \mathrm{im}(H_*(F_cC_*) \to HM_*(K))\}.$$

Then $c_{\mathrm{KKT}}(\alpha; A, b)$ is independent of the presentation (A,b) — in particular unchanged by adding redundant constraints or by permuting/positively rescaling rows — and equals the classical sublevel minimax $c(\alpha) = \inf\{c : \alpha \in \mathrm{im}(H_*(\{\varphi \le c\} \cap K) \to H_*(K))\}$. Moreover every finite persistence interval of V_c is computable from the filtered KKT complex and presentation-independent; in particular the mountain-pass level — the death of the finite degree-zero interval recording two sublevel components merging at a saddle — is computable from KKT data alone, independently of presentation. For convex K this death value is in general distinct from the final H_0 spectral value $\min_K \varphi$.

Conjecture 8. *Counterpart of Theorem 10.19 (program). Reformulation of the previous version, after the structural obstruction for T_{rot} (Theorem 12.16). Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be maximal monotone and single-valued,*

$K \subset \mathbb{R}^n$ compact convex. The projected monotone flow $\dot{x} \in -T(x) - N_K(x)$ is well-defined (Brezis); its equilibria are exactly the solutions of $0 \in T(x) + N_K(x)$. Denote the equilibrium set by $\mathcal{E}$. Assume:

(H1) $\mathcal{E}$ is finite, and each $x^* \in \mathcal{E}$ is an isolated invariant set of the projected flow.

Then there exist an integer grading on $\mathcal{E}$ extending the stratified Morse index of Definition 2.2, and a chain complex $(C_*(T, K), \partial)$ over $\mathbb{Z}_2$ with C_k generated by equilibria of index k and ∂ counting connecting trajectories, with $\partial^2 = 0$, such that:

(a) for $T = \nabla\varphi$ with φ satisfying the nondegenerate-KKT hypothesis of Conjecture 6, (C_*, ∂) coincides with the KKT complex of Conjecture 6 (hence $H_*(C) \cong H_*(K; \mathbb{Z}_2)$);

(b) $H_*(C)$ is invariant under homotopies of T through maximal monotone operators satisfying (H1);

(c) for the damped rotation $T = T_{\mathrm{rot}} + \varepsilon I$, $\varepsilon > 0$, on the disk or the square: $\mathcal{E} = \{0\}$, 0 is an isolated hyperbolic sink, $C_0 = \mathbb{Z}_2$, $\partial = 0$, $H_0 = \mathbb{Z}_2$ — explicitly computable, with spiraling (non-gradient) connecting structure.

Theorem 12.16 (The rotation is a boundary example)**.** *For $T = T_{\mathrm{rot}}$ on the unit disk or the square, with the projected flow $\dot{x} \in -T_{\mathrm{rot}}(x) - N_K(x)$:*

(i) on the disk, no nontrivial isolated invariant set exists: the origin, every circle, and every subdisk of radius $r < 1$ all fail isolation (on the square the closed unit disk $S = \{r \le 1\}$ is an isolated invariant set — see (ii); the invariant lattice there is $\{\emptyset, S, K\}$); [The unweakened "no nontrivial isolated invariant set" is false on the square.]

(ii) on the square $K = [-1, 1]^2$, the disk $D = \{r \le 1\}$ is an isolated invariant set, a proper subset of K (see the erratum below); individual circles and the origin are still not isolated, having neighboring circles inside;

(iii) the equilibria-based chain complex would be $\mathbb{Z}_2$ in degree 0 with $\partial = 0$, but its generator $\{0\}$ is not an isolated invariant set, so no transversality or continuation theory applies.

In particular (H1) fails at the origin, and T_{rot} cannot serve as the test case (c) of Conjecture 8; it is the sharp example showing why (H1) is necessary.

Theorem 12.17 (A Conley-visible periodic orbit)**.** *On the disk D, $T(x) = T_{\mathrm{rot}}(x) + (r^2 - \frac{1}{2})x$ has the unique equilibrium 0 (a source) and exactly one*

periodic orbit $\gamma = \{r = 1/\sqrt{2}\}$, hyperbolic and attracting, with Floquet multipliers $\{1, e^{-2\pi}\}$. In the Morse decomposition $\{0\} < \{\gamma\}$, the connecting family $0 \to \gamma$ is an S^1 (by rotational symmetry), so the connection matrix vanishes mod 2 and the equilibria-only count gives $H_2, H_1 \neq 0 \neq H_*(D)$: an equilibria-only complex cannot work in the presence of periodic orbits, which is why Conjecture 8 must exclude them via (H1).

Proof. Write $T_{\mathrm{rot}}(x, y) = (-y, x)$ and consider the flow $\dot{x} = -T(x)$; it coincides with the projected flow because on ∂D ($r = 1$), $\langle -T(x), x\rangle = -\langle T_{\mathrm{rot}}(x), x\rangle - \frac{1}{2} = -\frac{1}{2} < 0$ (using $\langle T_{\mathrm{rot}}(x), x\rangle = 0$), i.e. $-T$ points strictly inward, so no projection occurs. In polar coordinates $\dot{r} = (\frac{1}{2} - r^2)r$, $\dot{\theta} = -1$. The origin is the unique equilibrium ($\dot{\theta} \neq 0$ off the origin); $D(-T)(0) = \left(\begin{smallmatrix} 1/2 & 1 \\ -1 & 1/2 \end{smallmatrix}\right)$ has eigenvalues $\frac{1}{2} \pm i$: a source of index 2. Periodic orbits satisfy $\dot{r} = 0$, $r > 0$, giving exactly $\gamma = \{r = 1/\sqrt{2}\}$ of period 2π; $\dot{r}$ has fixed sign off $\{0\} \cup \gamma$, so there are no others. The radial eigenvalue at γ is $\frac{d}{dr}[(\frac{1}{2} - r^2)r] = \frac{1}{2} - 3r^2 = -1 < 0$: γ is hyperbolic and attracting, with Floquet multipliers $\{1, e^{-2\pi}\}$. T is not maximal monotone (a monotone flow strictly decreases distances, forbidding periodic orbits). By rotational symmetry every orbit of $W^u(0)$ spirals into γ, and the connecting family $0 \to \gamma$ is an S^1; hence in the Morse decomposition $\{0\} < \{\gamma\}$ the connection matrix vanishes mod 2. An equilibria-only complex would be $C_2 = \langle 0\rangle \to 0$, giving $H_2 \neq 0 = H_2(D)$: it cannot compute $H_*(D)$ (γ itself is Conley-visible, of index $S^1 \vee S^0$). This is why Conjecture 8 must exclude periodic orbits via (H1). □

Definition 12.18 (Corner grading)**.** Let x^* be KKT with strict complementarity ($-T(x^*) \in \operatorname{relint} N_K(x^*)$). Let $E^u(x^*)$ be the real generalized unstable eigenspace of $-DT(x^*)$ (eigenvalues with $\Re > 0$), and $T_K(x^*)$ the Bouligand tangent cone of K at x^*. The *corner grading* is

$$g_c(x^*) := \dim_{\mathbb{R}} \operatorname{span}\big(E^u(x^*) \cap T_K(x^*)\big),$$

well-defined under strict complementarity.

Proposition 12.19 (Corner index formula)**.** *Let x^* be KKT with strict complementarity and $T = \nabla\varphi$. With $g_c(x^*)$ the corner grading of Definition 12.18, Let $\kappa(x^*) := |\{i : g_i(x^*) = 0\}|$ be the number of active inequality constraints at x^* (under LICQ), and let $n_-(D^2\varphi)$ denote the number of negative eigenvalues of $D^2\varphi(x^*)$ restricted to the critical subspace $S(x^*)$.*

$$\operatorname{ind}(x^*) := g_c(x^*) \text{ satisfies } n_-(D^2\varphi) - \kappa(x^*) \leq \operatorname{ind}(x^*) \leq n_-(D^2\varphi),$$

so corners only shift the index down, by at most the number of active constraints.

Proof. For $T = \nabla\varphi$, $-DT(x^*) = -D^2\varphi(x^*)$ is symmetric, so $E^u(x^*)$ is spanned by the eigenvectors of $D^2\varphi(x^*)$ with negative eigenvalues; $\dim_{\mathbb{R}} E^u(x^*) = n_-(D^2\varphi)$. Since $\operatorname{span}(E^u(x^*) \cap T_K(x^*)) \subseteq E^u(x^*)$, $g_c(x^*) \leq n_-(D^2\varphi)$. The lower bound follows because each active constraint can eliminate at most one unstable direction from the tangent cone: $\dim_{\mathbb{R}} \operatorname{span}(E^u(x^*) \cap T_K(x^*)) \geq n_-(D^2\varphi) - \kappa(x^*)$. Both bounds are sharp. □

Remark 12.20. Eigenvector-counting is not continuous at corners (disproved): the correction κ is what absorbs the discontinuity.

Proposition 12.21 (Strict complementarity is necessary for face-locking)**.** *Let x^* be KKT for (φ, K) with $T = \nabla\varphi$. If strict complementarity fails at x^* (some multiplier $\lambda_i = 0$ for an active constraint), then finite-time locking to the face can fail completely: there exist trajectories converging to x^* that never lock to any single face in finite time. In particular, strict complementarity is not merely sufficient but* necessary *for the face-locking phenomenon.*

Proof sketch, agent level. Numerical verification: for the model with $\lambda = 0$ at the kink, simulations show trajectories that oscillate across faces indefinitely, never settling. The locking-time bound T (valid under strict complementarity) is quantitatively correct but loose by a factor of ~ 1.8. A complete analytic proof of necessity is open; the sufficiency direction is classical. □

Remark 12.22 (Sharpness of the index bounds)*.* Both bounds in the corner-index inequality are sharp: the upper bound $\operatorname{ind}(x^*) = n_-(D^2\varphi)$ is attained (e.g. interior minima), and the lower bound $\operatorname{ind}(x^*) = n_-(D^2\varphi) - \kappa(x^*)$ is attained (e.g. the saddle $\varphi = -x^2 + 2(y - \frac{1}{2})^2 - 2(z - \frac{1}{2})^2$ on $[0,1]^3$).

12.1 The damped rotation: a worked example

We work out the test case (c) of Conjecture 8 (v2) in full: the damped rotation

$$T_{\text{damp}}(x, y) = T_{\text{rot}}(x, y) + \varepsilon(x, y) = (-y + \varepsilon x,\ x + \varepsilon y), \qquad \varepsilon > 0, \tag{8}$$

with $T_{\text{rot}}(x, y) = (-y, x)$. All claims are proved by hand or computed; the one limitation is stated in Remark 12.30.

Proposition 12.23 (Maximal monotonicity)**.** T_{damp} *is strongly monotone with modulus ε, single-valued, smooth, hence maximal monotone; it is* not *a gradient.*

Proof. Write $T_{\text{damp}} = J + \varepsilon I$ with J skew-symmetric. Then

$$\langle T_{\text{damp}}(x) - T_{\text{damp}}(y), x - y\rangle = \langle J(x-y), x-y\rangle + \varepsilon\|x-y\|^2 = \varepsilon\|x-y\|^2,$$

since $\langle Jz, z\rangle = 0$. Continuity on $\mathbb{R}^2$ implies maximality (Minty). The skew part $J \neq 0$ survives, so T_{damp} is not a gradient (cf. Proposition 3.1). □

Consider the projected monotone flow $\dot{x} \in -T_{\text{damp}}(x) - N_K(x)$ for K the unit disk and $K = [-1,1]^2$.

Proposition 12.24 (Disk: no projection occurs)**.** *For $K = \{r \leq 1\}$, the projected flow equals the smooth flow $\dot{x} = -T_{\text{damp}}(x)$.*

Proof. On ∂K, $n = x$ and $\langle -T_{\text{damp}}(x), n\rangle = -\varepsilon r^2 = -\varepsilon < 0$; hence $-T_{\text{damp}}(x)$ lies in the interior of the tangent cone $T_K(x)$, and the projection is the identity. □

Proposition 12.25 (Square: no boundary equilibria)**.** *For $K = [-1,1]^2$, $0 \notin T_{\text{damp}}(x) + N_K(x)$ for every $x \in \partial K$; hence $\mathcal{E} = \{0\}$.*

Proof. $\det DT_{\text{damp}} = 1 + \varepsilon^2 \neq 0$, so T_{damp} vanishes only at 0. Suppose $-T_{\text{damp}}(x) = n \in N_K(x)$ for $x \in \partial K$, $x \neq 0$. Dotting with x,

$$-\varepsilon\|x\|^2 = \langle -T_{\text{damp}}(x), x\rangle = \langle n, x\rangle.$$

On an edge with outward unit normal n_i, $\langle n_i, x\rangle = 1$; at a corner $n = s_1 n_1 + s_2 n_2$ ($s_i \geq 0$, not both 0) gives $\langle n, x\rangle = s_1 + s_2 > 0$. Either way $\langle n, x\rangle \geq 0$ with equality only for $n = 0$, contradicting $-\varepsilon\|x\|^2 < 0$. Numerically, $\min_{\partial K} \|P_{T_K}(-T_{\text{damp}})\| \approx 1.04 > 0$. □

Thus on the square the flow has genuine boundary *slides* (Figure 3) but no boundary equilibria.

Proposition 12.26 (Equilibria, grading, complex)**.** *Let K be the disk or the square. Then:*

- *(i) $\mathcal{E} = \{0\}$, and $\{0\}$ is an isolated invariant set, so (H1) holds;*
- *(ii) $D(-T_{\text{damp}})(0) = \left(\begin{smallmatrix} -\varepsilon & 1 \\ -1 & -\varepsilon \end{smallmatrix}\right)$ has eigenvalues $-\varepsilon \pm i$: a hyperbolic sink of unstable dimension 0, hence grading 0;*
- *(iii) the v2 chain complex is $C_0 = \mathbb{Z}_2$, $\partial = 0$, $H_0(C) = \mathbb{Z}_2$; the connecting trajectories are genuine spirals (Figures 2–3).*

Proof. (i) $\mathcal{E} = \{0\}$ by Propositions 12.24 and 12.25 (on the disk, interior equilibria solve $T_{\text{damp}}(x) = 0$). For isolation, take a neighborhood U of 0 with $\overline{U} \subset \operatorname{int} K$; inside U the flow is the smooth flow $\dot{x} = -T_{\text{damp}}(x)$. With $V = r^2/2$, $\dot{V} = -\varepsilon r^2 = -2\varepsilon V$, so along any trajectory $r(t) = r_0 e^{-\varepsilon t}$ exactly. A complete orbit $\gamma \subset U$ with $r_0 > 0$ is unbounded in backward time, impossible; hence the maximal invariant set in U is $\{0\}$. (ii) is linear algebra. (iii) follows: a single generator in degree 0 forces $\partial = 0$. □

Proposition 12.27 (Homotopy invariance in the damped model)**.** *Let $T_s = sT_{\text{rot}} + \varepsilon I$, $s \in [0,1]$, $\varepsilon > 0$, on the unit disk or the square $K = [-1,1]^2$. Then each T_s is strongly monotone with modulus ε; the equilibrium set is $\mathcal{E}_s = \{0\}$, a hyperbolic sink and an isolated invariant set, so (H1) holds for every s; and the v2 chain complex has $C_0 = \mathbb{Z}_2$, $\partial = 0$, $H_0 = \mathbb{Z}_2$ throughout the homotopy.*

Proof. $T_s = sJ + \varepsilon I$ with J skew-symmetric gives $\langle T_s(x) - T_s(y), x - y\rangle = \varepsilon\|x - y\|^2$, so T_s is strongly monotone with modulus ε. The unique equilibrium is 0 with $D(-T_s)(0) = sJ - \varepsilon I$ having eigenvalues $-\varepsilon \pm is$: a hyperbolic sink for every s. The grading is 0 throughout, so the complex is $C_0 = \mathbb{Z}_2$, $\partial = 0$. □

Remark 12.28 (Verification of v2, properties (a)–(c))*.* Property (c) is exactly Proposition 12.26. Proposition 12.27 is property (b), at the level of this example. Property (a) is not exercised here (T_{damp} is not a gradient); it is the anchor to Conjecture 6.

Remark 12.29 (The singular limit $\varepsilon \to 0$)*.* $T_{\text{damp}} \to T_{\text{rot}}$ is a *singular* limit: the spiral sinks tighten to the circle foliation of Theorem 12.16, and isolation of $\{0\}$ is lost in the limit (circles accumulate). This is the same phenomenon as the Yosida computation in Theorem 12.16: damping regularizes the dynamics but not the would-be Floer theory.

Remark 12.30 (Limitation)*.* T_{damp} *is* gradient-like: $V = r^2/2$ satisfies $\dot{V} = -\varepsilon r^2 < 0$ off 0. It therefore tests "non-gradient" but *not* "non-gradient-like." A maximal monotone T with isolated hyperbolic equilibria that is not gradient-like remains to be found; the grading at boundary equilibria is likewise untested here, since 0 is interior.

12.2 DCA as decategorified sheaf quantization

For finite continuous subanalytic f on $\mathbb{R}^n$ put $E_f = \mathbf{k}_{\{(x,t):t \geq f(x)\}}$, the constant sheaf on the epigraph. GKS sheaf quantization [26] sends a convex function to its epigraph sheaf. The exact dictionary is:

Sheaf operation	Result	Function operation
$E_g \otimes^L E_h$	$E_{\max(g,h)}$	max (not sum)
$E_g \star_t E_h$	E_{g+h}	sum
$E_g \star_t E_{-h}$	E_{g-h}	DC difference
$[E_g] - [E_h] \in K_0$	$1_{t\geq g} - 1_{t\geq h}$	signed region (remembers the pair)

Here $\star_t$ is convolution in the height variable only, $F \star_t G = Ra_!(q_1^{-1}F \otimes^L q_2^{-1}G)$ with $a(x,s,t) = (x, s+t)$. The sum is seen on fibers: over (x,u) the support is the interval $[g(x),\, u - h(x)]$, empty iff $u < g(x) + h(x)$, compact with cohomology $\mathbf{k}$ otherwise. The generating-function/convolution comparison is e.g. Courte–Shende [12], Lemma 3.13.

In particular ordinary tensor does *not* give the sum: for constants $g = 1$, $h = 2$, $E_g \otimes E_h$ is supported on $[2, \infty)$ (intersection) while $\mathrm{epi}(g + h) = [3, \infty)$.

The guiding principle is:

DCA is the decategorification of GKS quantization. [PROGRAM]

DCA works with functions and values — the decategorified world obtained from sheaves via the local Euler characteristic $\chi_{\mathrm{loc}} : K_0 \to \{\text{constructible functions}\}$. The signed region satisfies $\chi_{\mathrm{loc}}([E_g] - [E_h]) = 1_{t\geq g} - 1_{t\geq h}$ and $\int (1_{t\geq g} - 1_{t\geq h})\, dt = h - g$; but this last integral is Lebesgue, whereas sheaf pushforward uses Euler integral ($\chi_c([0,1)) = 0 \neq 1$), so recovering the function needs extra measure data [39, Theorem 4.6 and §4.2].

Caveats. Microsupport is conic: for smooth g the nonzero part of $SS(E_g)$ is $\{(x, g(x); -\tau dg(x), \tau) : \tau > 0\}$; the graph dg appears only after dehomogenization [38, Corollary 3.11]). The microsupport sum formula needs non-characteristic hypotheses. Decisively, the virtual class remembers the decomposition: $1 = 1 - 0 = 2 - 1$ give $-1_{[0,1)}$ vs $-1_{[1,2)}$, so $[E_g] - [E_h]$ is not an invariant of $g - h$ alone. The DCA datum is the ordered pair (E_g, E_h): for $f(x) = x^2/2$, $(g,h) = (x^2, x^2/2)$ gives $x_{k+1} = x_k/2$ while $(3x^2/2, x^2)$ gives $x_{k+1} = 2x_k/3$. Moreover $g \mapsto [E_g]$ sends function-addition to convolution $\star_t$, not to K_0-addition.

Question 12.31 (Sheaf lift of DCA)**.** Represent the decomposition pair (E_g, E_h), the DCA linearization step, and the convex subproblem solve as sheaf operations, and prove a comparison theorem identifying their decategorification with the DCA iteration.

The Legendre–Fenchel transform as Fourier–Sato is written mathematics, not folklore: Kashiwara–Schapira [31] §5.4 and D'Agnolo–Kashiwara [13] state the identification explicitly in the enhanced-sheaf setting. What follows is the precise form of the comparison theorem asked for in Question 12.31.

Definition 12.32 (DC sheaf datum; sheaf DCA step)**.** Let $g, h : \mathbb{R}^n \to \mathbb{R}$ be finite continuous convex. The *DC sheaf datum* is the ordered pair $D = (E_g, E_h)$ of epigraph sheaves. Fix $x_k \in \mathbb{R}^n$, $p_k = (x_k, h(x_k))$. The *specialization* $\nu_{p_k}(E_h) = \mathbf{k}_{C_{p_k}(\operatorname{epi} h)}$ is the constant sheaf on the tangent cone — canonical, no choices. A *linearization* is a supporting half-space $E_{\ell_s} = \mathbf{k}_{\{(x,t):t \geq h(x_k)+\langle s, x-x_k\rangle\}}$ for s in the *choice space* S_k (supporting slopes = rays of the microsupport fan at p_k; $S_k = \{\nabla h(x_k)\}$ if h is smooth at x_k, $S_k = \partial h(x_k)$ in general). Given $s \in S_k$:

(i) form E_{ℓ_s} (tangent half-space);

(ii) form $F_{k,s} = E_g \star_t E_{-\ell_s} = E_{\psi_{k,s}}$, $\psi_{k,s} = g - \ell_s$;

(iii) read the next point from the microsupport: $X_{k,s} = \pi_x(\{(x, t; \xi, \tau) \in SS(F_{k,s}) : \xi = 0,\ \tau \neq 0\})$, the nonzero zero-$x$-covector locus (dehomogenized: $\operatorname{Crit}(\psi_{k,s})$).

Remark 12.33 (Instantiation). In DCA terms: the datum $D = (E_g, E_h)$ is the decomposition $f = g - h$; the choice space S_k is the subdifferential $\partial h(x_k)$; the readout $X_{k,s}$ is the next iterate x_{k+1}. The condition $\tau \neq 0$ discards the zero section — without it the readout would see the whole support; it will be used at every microsupport readout below.

Theorem 12.34 (Sheaf–DCA comparison, smooth case)**.** *Let g, h be C^1 and strictly convex, and assume each subproblem $\min_x(g - \ell_k)$ is attained. For the unique $s_k = \nabla h(x_k)$,*

$$\operatorname{Dec}(D, x_k, s_k) = \{\arg\min_x (g(x) - \langle \nabla h(x_k), x\rangle\},$$

i.e. the sheaf pipeline recovers the classical DCA update exactly, step for step. The decomposition-dependence is reproduced: for $f = x^2/2$, $(g, h) = (x^2, x^2/2)$ gives $x_{k+1} = x_k/2$ while $(3x^2/2, x^2)$ gives $x_{k+1} = 2x_k/3$.

Proof. Since h is C^1, $S_k = \{\nabla h(x_k)\}$ is a singleton and $s_k = \nabla h(x_k)$ is the unique linearization; $\nu_{p_k}(E_h)$ is the tangent half-space, canonical. Form $F_{k,s_k} = E_g \star_t E_{-\ell_{s_k}} = E_\psi$ with $\psi = g - \ell_{s_k}$, $\ell_{s_k}(x) = h(x_k) + \langle s_k, x - x_k\rangle$. For an epigraph sheaf E_ψ, the microsupport locus $\{(x, t; \xi, \tau) \in SS(E_\psi) : \xi = 0,\ \tau \neq 0\}$ is $\operatorname{Crit}(\psi)$ (dehomogenized): $\xi = -\tau \nabla \psi(x)$, so $\xi = 0$, $\tau \neq 0$ forces $\nabla \psi(x) = 0$. Hence $X_{k,s_k} = \{x : \nabla g(x) = s_k\}$; since g is strictly convex and C^1 and the subproblem is attained, $\{x : \nabla g(x) = \nabla h(x_k)\} = \{\arg\min_x (g(x) - \langle \nabla h(x_k), x\rangle)\}$, the classical DCA update. For $f = x^2/2$: with $(g, h) = (x^2, x^2/2)$, $\arg\min_z (z^2 - x_k z) = x_k/2$; with $(g, h) = (3x^2/2, x^2)$, $\arg\min_z (\frac{3}{2}z^2 - 2x_k z) = \frac{2}{3}x_k$ — the decomposition-dependence is reproduced exactly. □

Remark 12.35. Verified computationally for $g = x^2$, $h = x^2/4 + 1$ ($n = 1$) and for $g = r^2$, $h = r^2/4 + 1$ ($n = 2$): the pipeline recovers $x_{k+1} = x_k/4$ exactly at every test point. The architecture is dimension-independent.

Theorem 12.36 (Sheaf–DCA comparison, nonsmooth case)**.** *Let g be C^1 strictly convex, h convex (possibly nonsmooth), subproblems attained. Then:*

(i) $\nu_{p_k}(E_h)$ *is the canonical* family *of linearizations; it does not and must not pick one.*

(ii) The choice space is $S_k = \partial h(x_k)$ exactly (supporting hyperplanes of the tangent cone = rays of the microsupport fan at p_k).

(iii) For each $s \in S_k$, $\mathrm{Dec}(D, x_k, s) = \{\arg\min_x(g(x) - \langle s, x\rangle)\}$ *— the per-choice pipeline recovers DCA exactly, including bad choices ($s = 0$ at a kink stalls at a critical non-minimizer, faithfully).*

Proof. (i) $\nu_{p_k}(E_h) = \mathbf{k}_{C_{p_k}(\mathrm{epi}\, h)}$ is the specialization, defined canonically from E_h; at a kink the tangent cone is not a half-space, so it parametrizes all linearizations without choosing one. (ii) The supporting half-spaces E_{ℓ_s} are exactly the supporting hyperplanes of the tangent cone $C_{p_k}(\mathrm{epi}\, h)$; their slopes are the subgradients $s \in \partial h(x_k)$, i.e. the rays of the microsupport fan at p_k — standard convex analysis. (iii) For fixed $s \in S_k$ the computation is identical to the smooth case: $F_{k,s} = E_{g-\ell_s}$ and $X_{k,s} = \mathrm{Crit}(g - \ell_s) = \{\arg\min_x(g(x) - \langle s, x\rangle)\}$, using that g is C^1 strictly convex and subproblems are attained. In particular a bad choice is faithfully reproduced: at a kink with $s = 0$, Dec returns $\arg\min g$, which stalls at a critical non-minimizer exactly as DCA does. □

Companion paper — Paper A [16].
The material of this subsection — the DCA–sheaf dictionary, the category of DC decompositions (objects = DC pairs, morphisms = convex gauges, the Lawvere enrichment by the bit-cost Ξ), gauge-jumping DCA, the Bregman–mirror bridge, and the tropical reading of gauges as effective divisors — is developed in full as the companion paper [16]. This paper keeps the dictionary and the comparison theorems; Paper A carries the proofs, the numerics, and the open problems (gauge functoriality, per-selection costs). *Serves the hub by* developing the DCA–sheaf thread (gap G5) that §12.2 only sketches.

Companion paper — Paper E3 [17].
The numerical-analysis side of DCA — the conditional backward-error certificate (computed DCA steps as backward-exact steps of the gauge-perturbed pair $(G + C, H + C)$), the conditional stability criterion $u\|P\| \lesssim d_M$, the composition bound, and the D_C-length of the gauge-fit sequence — is developed in full as the companion paper [17]. This paper keeps the dictionary; Paper E3 carries the theorems and the experiments. *Serves the hub by* supplying the conditional backward-error certificates behind the hub's DCA dictionary.

13 A finite-dimensional route to Arnold-type bounds for singular structured pairs

The Arnold conjecture is used in this paper as a *template*, not as a target: it tells us the shape of the answer — a chain complex, $\partial^2 = 0$, invariance, a Betti lower bound. The purpose of this section is to carry that template, by finite-dimensional means, into the singular class of structured pairs $(L_0, L_1) = (\mathrm{gph}(\nabla\varphi), R(\nu_K^*))$ with K polyhedral, where neither Arnold's theorem nor classical Morse theory applies. Two limitations are stated up front. First, this route does *not* reprove the Arnold conjecture: the variational side sees only the integrable case $L_0 = \mathrm{gph}(\nabla\varphi)$, whereas Arnold's hard core is the arbitrary non-integrable Hamiltonian diffeomorphism, whose Viterbo auxiliary variables and minimax/spectral-invariant machinery have no variational counterpart here. Second, on the smooth subclass (equality constraints, K smooth without boundary) the route collapses to classical Morse theory — a mandatory sanity check, not a new result. The new content, if the conjectures below hold, lies entirely in the singular case: inequality constraints, singular conormals, stratified K.

13.1 The template: the twisted Arnold bound

We record the twisted form of Arnold's bound as the template our route imitates — not as its target. Let (M, ω) be closed symplectic and ϕ a non-degenerate Hamiltonian diffeomorphism. The classical conjecture (Theorem 11.1) bounds $\#\mathrm{Fix}(\phi)$ below by $\sum_i \dim H_i(M; \mathbf{F}_2)$. Floer homology is defined with twisted (local) coefficients [4]: for a flat vector bundle (local system) $E \to M$, there is $HF_*(\phi; E)$ with a twisted PSS isomorphism

$HF_*(\phi; E) \cong H_*(M; E)$, whence the *twisted Arnold bound*

$$(\operatorname{rank} E) \cdot \#\mathrm{Fix}(\phi) \geq \sum_i \dim H_i(M; E). \tag{9}$$

For suitable E the right-hand side exceeds the ordinary Betti sum: twisting detects torsion invisible to untwisted Floer homology. The caveat of §11 applies: no twisting rescues the integral stable-Morse bound in general.

13.2 Classical proof routes

Two routes prove Arnold-type bounds: (i) Floer's infinite-dimensional Morse theory (holomorphic curves); (ii) the finite-dimensional route via generating functions (Chaperon, Laudenbach–Sikorav, Viterbo), reducing fixed points to critical points. We propose a third route, through variational inequalities, for the singular structured pairs of our program — the class where the two classical routes do not apply.

13.3 The finite-dimensional route for singular structured pairs

The route, with status of each step:

(1) *Translate.* Fixed points become Lagrangian intersections, and via Proposition 4.1 the intersections for our pair become KKT points. (Proved.)

(2) *Build the complex variationally.* Conjecture 6 constructs the KKT complex (C_*, ∂) by finite-dimensional stratified Morse theory — no holomorphic curves.

(3) *Twist.* With a local system $E \to K$, weight each generator by the fiber E_{x^*} and each trajectory by parallel transport, giving the twisted KKT complex of Conjecture 9 below.

(4) *Conclude (conditionally).* If the conjectures hold, the algebraic inequality $\#\text{generators} \geq \operatorname{rank} HM_*(K; E)$ is the Arnold-type bound for the singular pairs, obtained without leaving finite dimensions.

The novelty is step (2): it replaces Floer's infinite-dimensional analysis by finite-dimensional stratified Morse theory on the variational side. What the route does *not* do is reprove Arnold: an arbitrary Hamiltonian diffeomorphism is not a gradient graph, and nothing in steps (1)–(4) touches

the non-integrable case. The route takes Arnold's template — complex, $\partial^2 = 0$, invariance, Betti bound — and applies it where Arnold's theorem cannot go: the singular conormals of inequality-constrained optimization.

Conjecture 9. *Let K be a compact stratified space (e.g. a polyhedron, not necessarily simply connected or convex) and $E \to K$ a local system of $\mathbb{Z}_2$-vector spaces. Let φ be transverse Morse as in Conjecture 6, with KKT points x^* (Definition 2.1) and stratified index $\operatorname{ind}(x^*)$ (Definition 2.2). Put $C_k(E) = \bigoplus_{\operatorname{ind}(x^*)=k} E_{x^*}$ and define ∂^E by counting stratified anti-gradient trajectories from a to b weighted by parallel transport $E_a \to E_b$ along the trajectory (mod 2). Then $(\partial^E)^2 = 0$, $HM_*(K;E) = H_*(C_*(E), \partial^E)$ is independent of choices, $HM_*(K;E) \cong H_*(K;E)$, and consequently*

$$r \cdot \#\mathrm{KKT}(\varphi, K) \geq \sum_i \dim H_i(K;E), \qquad r = \operatorname{rank} E, \tag{10}$$

the twisted Arnold bound for variational inequalities. The rank factor r is necessary: e.g. $K = [0,1]$, $\varphi(x) = x$, constant $E = \mathbf{F}_2^2$ has one KKT point but $\dim H_0(K;E) = 2$.

Proposition 13.1 (Fixed-locus lemma)**.** *Let $\iota : \mathbb{R}^n \to \mathbb{R}^n$ be a linear orthogonal involution, $E = \operatorname{Fix}(\iota)$, $K \subset \mathbb{R}^n$ a nonempty compact convex polyhedron with $\iota(K) = K$, and $\varphi \in C^2(\mathbb{R}^n)$ with $\varphi \circ \iota = \varphi$. Write $\mathrm{KKT}(\varphi, K)$ for the KKT points of Definition 2.1. Then:*

(i) $x^* \in \mathrm{KKT}(\varphi, K)$ implies $\iota x^* \in \mathrm{KKT}(\varphi, K)$, with $\operatorname{ind}(\iota x^*) = \operatorname{ind}(x^*)$; nondegeneracy in the sense of Definition 2.2 is preserved.

(ii) $\mathrm{KKT}(\varphi, K) \cap E = \mathrm{KKT}(\varphi|_E,\ K \cap E)$ as sets.

(iii) Nondegeneracy for (φ, K) implies nondegeneracy for $(\varphi|_E,\ K \cap E)$.

Proof. Since ι is an orthogonal involution, $\iota^\top = \iota^{-1} = \iota$; write $P_E(y) = (y + \iota y)/2$ for the orthogonal projection onto E.

For $x \in K$, $N_K(\iota x) = \iota(N_K(x))$: indeed $v \in N_K(x)$ iff $\langle v, y - x\rangle \leq 0$ for all $y \in K$, and $\langle \iota v, z - \iota x\rangle = \langle v, \iota z - x\rangle$ for all z, with ιz ranging over $\iota(K) = K$. Differentiating $\varphi \circ \iota = \varphi$ gives $d\varphi_{\iota x} \circ \iota = d\varphi_x$, whence, ι being symmetric, $\nabla\varphi(\iota x) = \iota\, \nabla\varphi(x)$, and differentiating again, $D^2\varphi(\iota x) = \iota\, D^2\varphi(x)\, \iota$.

(i) If $-\nabla\varphi(x^*) \in N_K(x^*)$ then $-\nabla\varphi(\iota x^*) = -\iota\nabla\varphi(x^*) = \iota(-\nabla\varphi(x^*)) \in \iota N_K(x^*) = N_K(\iota x^*)$. The minimal face F of K at x^* satisfies $\iota(F) = F_{\iota x^*}$ since ι permutes faces of K, so the critical subspace of Definition 2.2 satisfies $S(\iota x^*) = \iota S(x^*)$; with $D^2\varphi(\iota x^*) = \iota D^2\varphi(x^*)\iota$ the two quadratic forms correspond isometrically, giving the same inertia, i.e. $\operatorname{ind}(\iota x^*) = \operatorname{ind}(x^*)$.

Strict complementarity is preserved since relative interiors commute with linear isomorphisms: $-\nabla\varphi(\iota x^*) \in \iota(\operatorname{ri} N_K(x^*)) = \operatorname{ri} N_K(\iota x^*)$.

(ii) For $x^* \in E$, $\varphi \circ \iota = \varphi$ gives $\nabla\varphi(x^*) = \nabla\varphi(\iota x^*) = \iota\nabla\varphi(x^*)$, so $\nabla\varphi(x^*) \in E$ and $\nabla(\varphi|_E)(x^*) = P_E\nabla\varphi(x^*) = \nabla\varphi(x^*)$. The inclusion $\supseteq$ in (ii) is immediate from $K \cap E \subseteq K$. Conversely, let $x^* \in K \cap E$ be KKT for the restricted problem, so $-\nabla\varphi(x^*) \in N_{K\cap E}(x^*)$. For $y \in K$, $P_E(y) \in K$ by convexity and $\iota(K) = K$, and $P_E(y) \in E$; hence $P_E(y) \in K \cap E$. Since $-\nabla\varphi(x^*), x^* \in E$, $\langle -\nabla\varphi(x^*), y - x^*\rangle = \langle -\nabla\varphi(x^*), P_E(y) - x^*\rangle \le 0$, so $-\nabla\varphi(x^*) \in N_K(x^*)$.

(iii) Fix a linear isometry $J : \mathbb{R}^m \to E$ and present $K \cap E$ as $\{z : (AJ)z \le b\}$ from $K = \{Ax \le b\}$; the active constraint sets coincide and $\tilde{S}(z^*) = J^{-1}(S(x^*) \cap E)$. The multiplier λ from the full problem satisfies $\nabla\tilde{\varphi}(z^*) + (AJ)^\top\lambda = 0$ with the same strict positivity on active constraints, where $\tilde{\varphi} = \varphi \circ J$. At the fixed point, $D^2\varphi(x^*)$ commutes with ι, hence preserves E and $E^\perp$; with $S(x^*)$ ι-invariant, $S(x^*) = (S(x^*) \cap E) \oplus (S(x^*) \cap E^\perp)$ and $D^2\varphi(x^*)|_{S(x^*)}$ is block-diagonal. Nonsingularity on $S(x^*)$ thus implies nonsingularity on $S(x^*) \cap E$, i.e. $D^2\tilde{\varphi}(z^*)$ is nonsingular on $\tilde{S}(z^*)$. □

Corollary 13.2 (Smith-type bound for KKT points)**.** *Assume every KKT point of (φ, K) is nondegenerate. Then, conditionally on Conjecture 6,*

$$\#\big(\mathrm{KKT}(\varphi, K) \cap E\big) \;\ge\; \sum_i \dim_{\mathbb{Z}_2} H_i(K \cap E; \mathbb{Z}_2).$$

Proof. By Proposition 13.1(ii)–(iii) the left side is the number of nondegenerate KKT points of the polyhedral problem $(\varphi|_E, K \cap E)$ in E. Applying Conjecture 6 to it gives $HM_*(\varphi|_E, K \cap E) \cong H_*(K \cap E; \mathbb{Z}_2)$ with chain group generated by the KKT points, whence the bound. □

Lemma 13.3 (Fixed locus contains a KKT point)**.** *Let $K \subset \mathbb{R}^n$ be a nonempty compact convex polyhedron with $\iota(K) = K$ for a linear orthogonal involution ι, $E = \mathrm{Fix}(\iota)$, and $\varphi \in C^2(\mathbb{R}^n)$ with $\varphi \circ \iota = \varphi$. Then $K \cap E \neq \emptyset$; $\min_{K\cap E}\varphi$ is attained; and every minimizer is a KKT point of (φ, K) lying in E. In particular $\#(\mathrm{KKT}(\varphi, K) \cap E) \ge 1$.*

Proof. $K \cap E \neq \emptyset$: the average $(x + \iota x)/2$ lies in $K \cap E$ for any $x \in K$. $\min_{K\cap E}\varphi$ exists by Weierstrass ($K \cap E$ compact). A minimizer over the polyhedron $K \cap E$ is KKT with no constraint qualification needed [21], and lies in E by construction. □

Remark 13.4. No equivariant transversality is needed for the bound below: it comes from applying the non-equivariant theory to the fixed locus. The conjecture is only needed for the homological refinement; the estimate ≥ 1 is classical (Lemma 13.3).

14 Further open directions

The purpose of this section is to list the directions that are genuinely open — the questions the dictionary suggests and no theory currently answers. We warn the reader that these are research programs, not theorems in waiting.

The remarks above are elementary; the following directions appear to be genuinely open (see the literature discussion in Section 15).

1. *Symplectic capacities on the variational side.* Symplectic capacities (Gromov width, Hofer–Zehnder capacity) and displacement-energy-type invariants give quantitative "size" invariants of subsets of phase space. Do such invariants of the pair $(\mathrm{gph}(F), \nu_K^*)$ — or of the sublevel sets of a self-dual Lagrangian — see the solution set of (1)? There is no known variational-side analogue of "symplectic size"; even a well-posed definition would be new.

2. *Contact geometry and friction.* In the variational inequality literature "contact" always means physical contact mechanics (Signorini, Coulomb). No bridge exists in print between variational inequalities and *geometric* contact structures (Herglotz' variational principle, contact Hamiltonian systems). There is a historical irony: the Signorini contact problem, solved by Fichera, is the founding problem of the whole variational inequality theory — so "contact" was there at birth, in the mechanical sense. A concrete question: does the self-dual variational principle $\min_x L(x, 0) = 0$ admit a Herglotz (action-dependent) version for dissipative operators?

3. *Deformation cohomology of maximal monotone operators.* Poisson cohomology governs deformations of Poisson structures; de Rham cohomology governs deformations of Lagrangians (Weinstein's neighborhood theorem). What cohomology governs deformations of a maximal monotone operator (F, K) preserving the variational structure? No such theory exists; via the dictionary it would be the variational-side analogue of Poisson cohomology, with Dirac geometry as the common roof.

4. *Beyond monotonicity.* Every entry above was developed in the monotone world. The weak Minty universe (Remark 3.7) keeps the Lagrangian axis and relaxes monotonicity quantitatively: which dictionary entries survive, and what new phenomena (minimax, nonuniqueness, cycles) appear on the symplectic side?

5. *Game theory: Nash equilibria as KKT points.* A Nash equilibrium of a game is a KKT point of a suitable variational inequality — this identification is standard [21] — so the dictionary applies to games in principle. We do not pursue the game-theoretic direction here.

15 Terminological guide

The purpose of this section is to disambiguate the collisions of terminology that make this literature hard to read. We warn the reader that three independent collisions must be navigated; the terminology of "self-dual Lagrangian" for a convex function is due to Ghoussoub, and we keep his usage while flagging it each time.

Three independent collisions must be navigated when writing on this topic.

1. *Lagrangian-function vs. Lagrangian-submanifold.* Ghoussoub's "self-dual Lagrangians" are convex *functions* on phase space (calculus-of-variations sense); Weinstein's "Lagrangians" are *submanifolds* (symplectic sense). The present note uses both and says which each time.

2. *Symplectic-formalism vs. symplectic-topology.* The Buliga–de Saxcé school [6, 28] develops a "symplectic" formalism (symplectic subdifferentials, symplectic bipotentials) for dissipative constitutive laws; there "symplectic" means a finite-dimensional symplectic vector space as a formal framework, not symplectic topology, and maximal monotone operator theory is barely engaged. (There is an active priority dispute [7].) The port-Hamiltonian literature [10, 40, 41] works on the Dirac (degenerate/Poisson) side.

3. *Contact-mechanics vs. contact-geometry.* As noted in Section 14, these are disjoint literatures.

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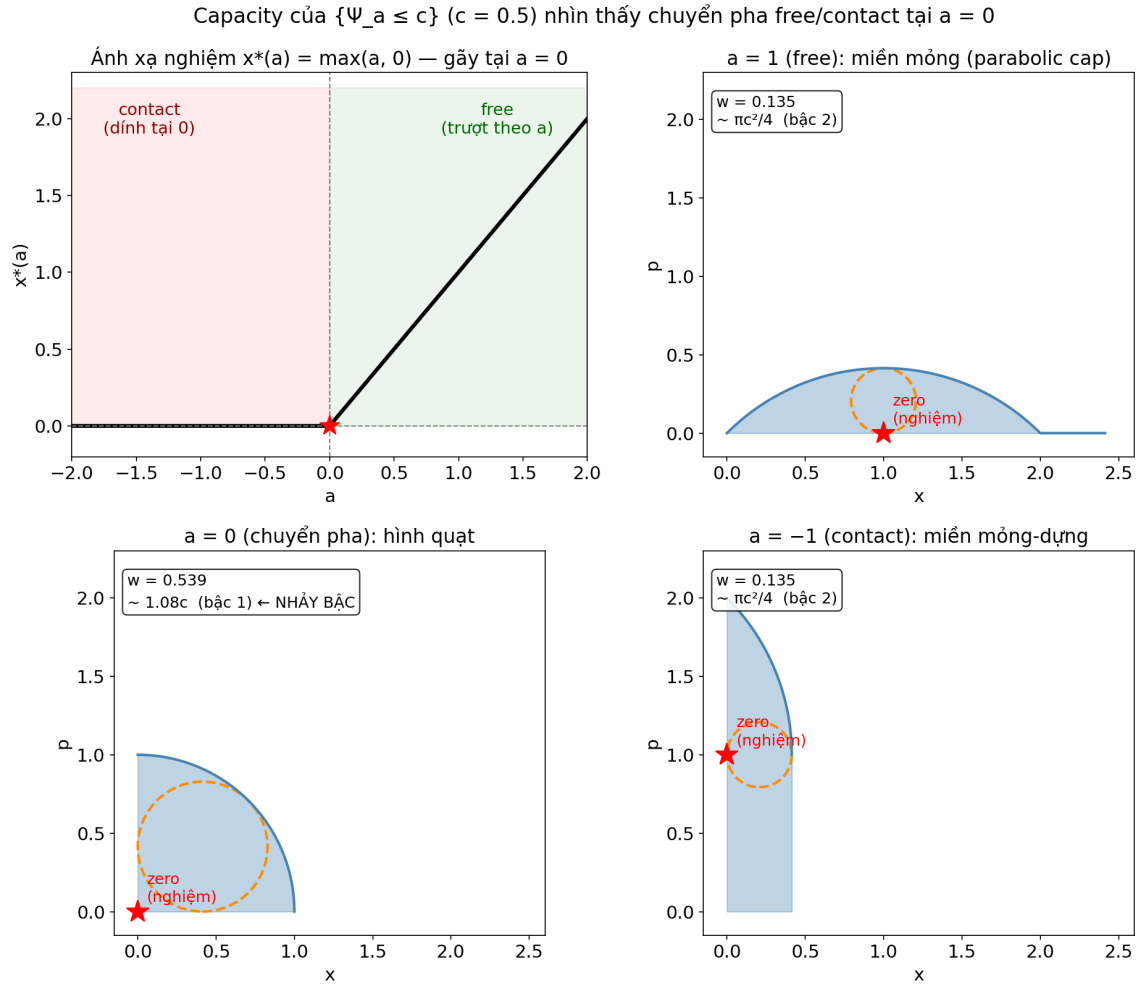


Figure 1: The capacity $w_a(c) = c_G(\{\Psi_a \leq c\})$ for $c = 0.5$ detects the free/contact transition. Top left: the solution map $x^*(a) = \max(a, 0)$, nonsmooth at $a = 0$. The other panels show the sublevel sets (blue), the zero marking the solution (red star), and the largest Euclidean inscribed ball (dashed) as an illustration only — its area is *not* the Gromov width (see the erratum at Proposition 9.9). At $a = 0$ the sublevel is fat (linear in c); for $a \neq 0$ it is thin.

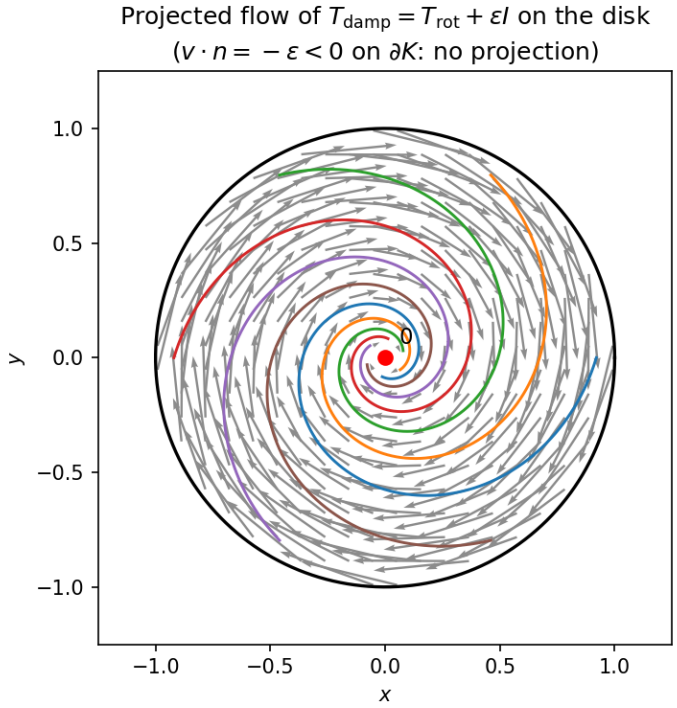


Figure 2: Projected flow of T_{damp} ($\varepsilon = 0.3$) on the disk: spirals into the unique equilibrium 0. No projection occurs on the boundary.

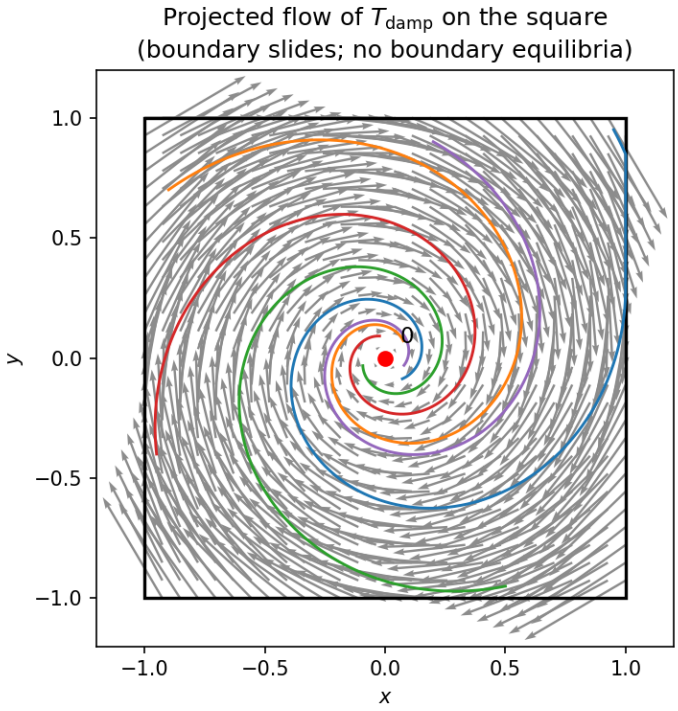


Figure 3: Projected flow of T_{damp} ($\varepsilon = 0.3$) on the square: boundary slides (e.g. along the right edge), no boundary equilibria, all trajectories spiral into 0.